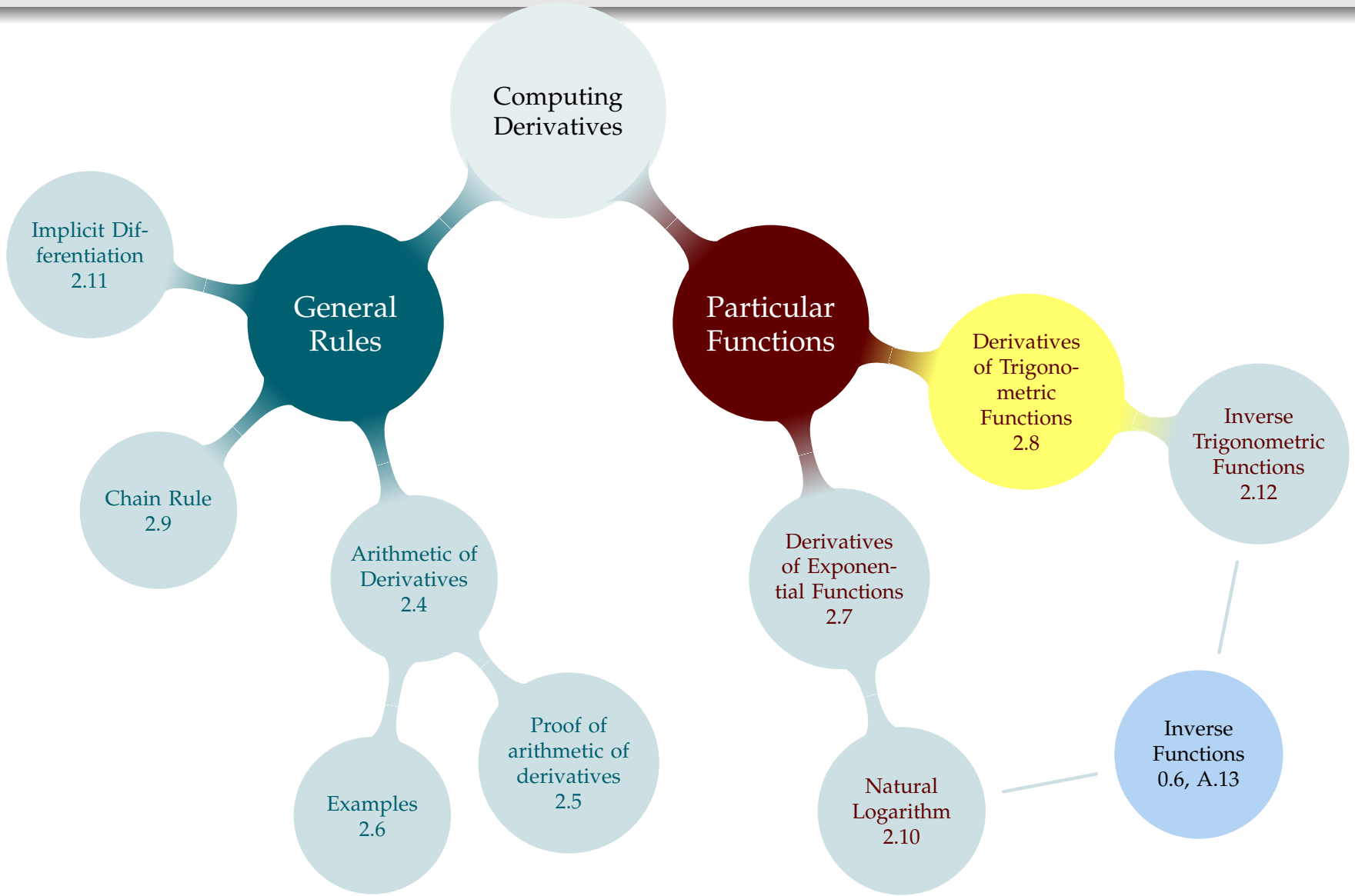
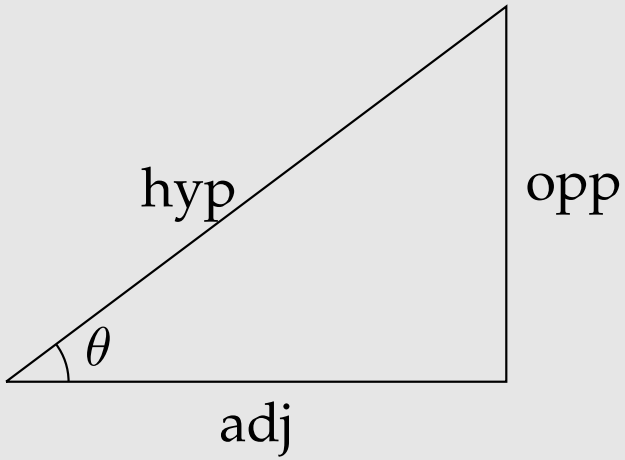
Table of Contents

SKIP TRIG REVIEW



Basic Trig Functions



$$\sin(\theta) = \frac{\text{opp}}{\text{hyp}} \qquad \csc(\theta) = \frac{1}{\sin(\theta)}$$

$$\cos(\theta) = \frac{\text{adj}}{\text{hyp}} \qquad \sec(\theta) = \frac{1}{\cos(\theta)}$$

$$\tan(\theta) = \frac{\text{opp}}{\text{adj}} \qquad \cot(\theta) = \frac{1}{\tan(\theta)}$$

Commonly used facts

- Graphs of sine, cosine, tangent

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- Sine, cosine, and tangent of reference angles: 0, $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$

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- How to use reference angles to find sine, cosine and tangent of other angles

Commonly used facts

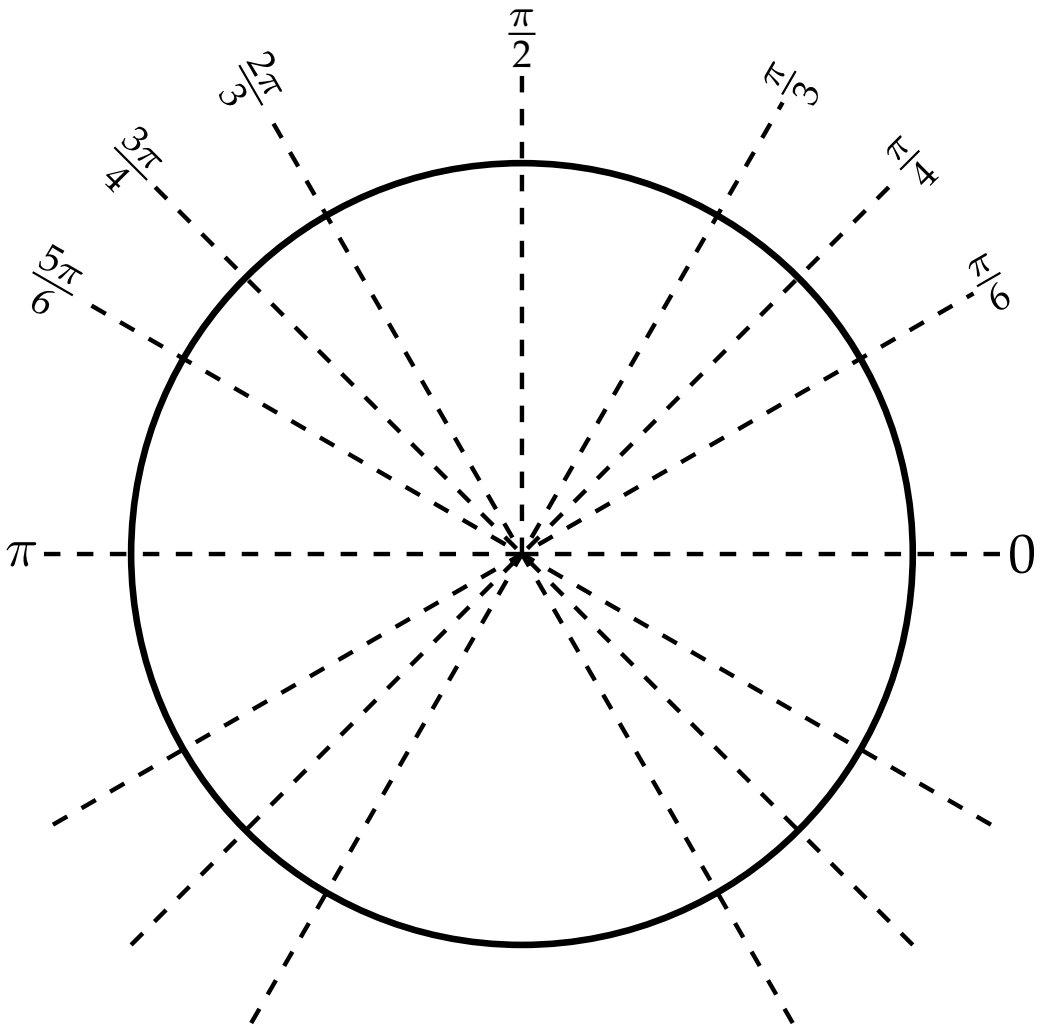
- Graphs of sine, cosine, tangent
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COMMONLY USED FACTS

- Graphs of sine, cosine, tangent
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- Conversion between radians and degrees

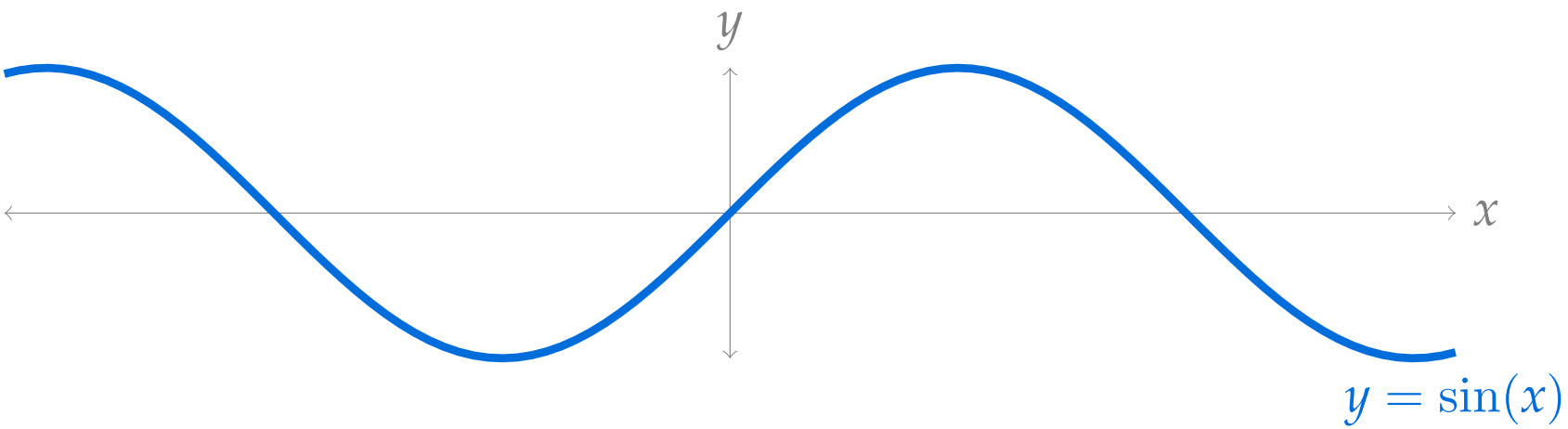
CLP-1 has an appendix on high school trigonometry that you should be familiar with.

Reference Angles



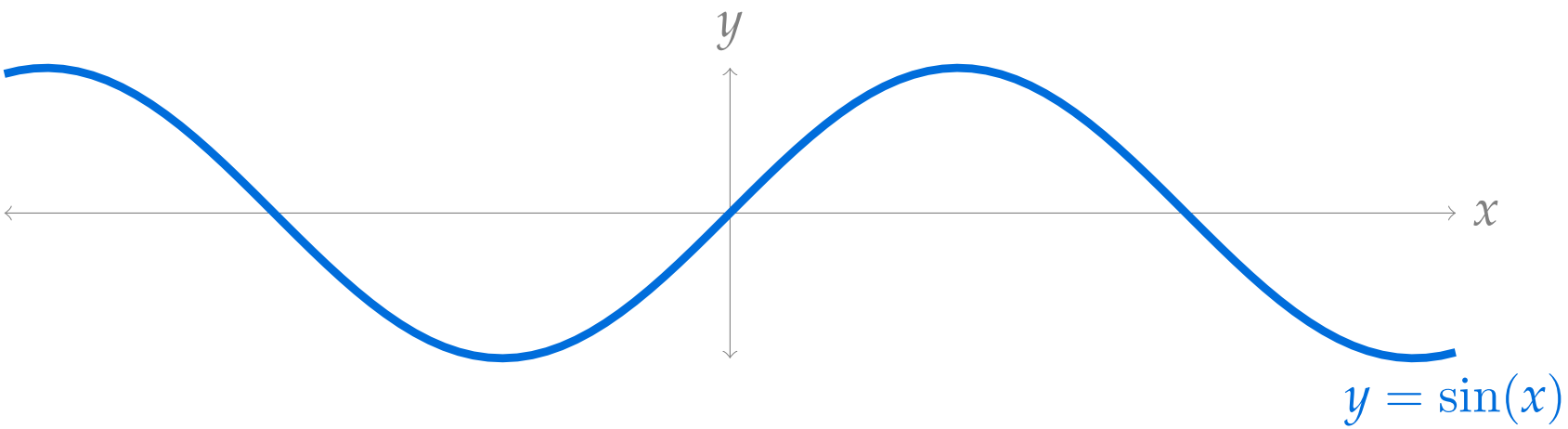
Derivative of Sine

SKIP PROOFS OF SINE AND COSINE DERIVATIVES



Derivative of Sine

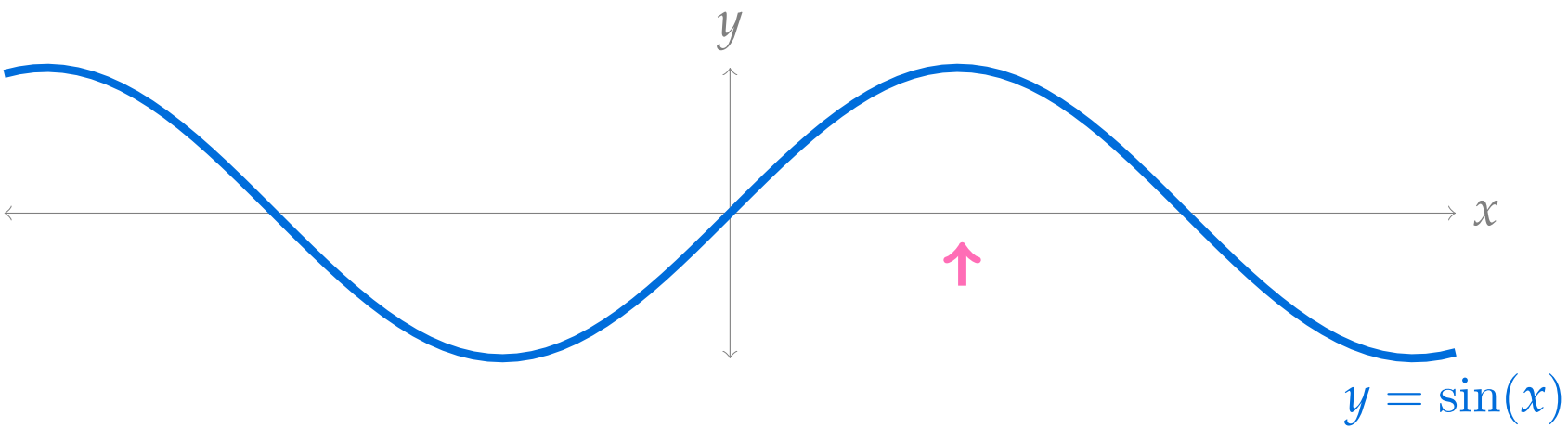
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Consider the derivative of $f(x) = \sin(x)$.

Derivative of Sine

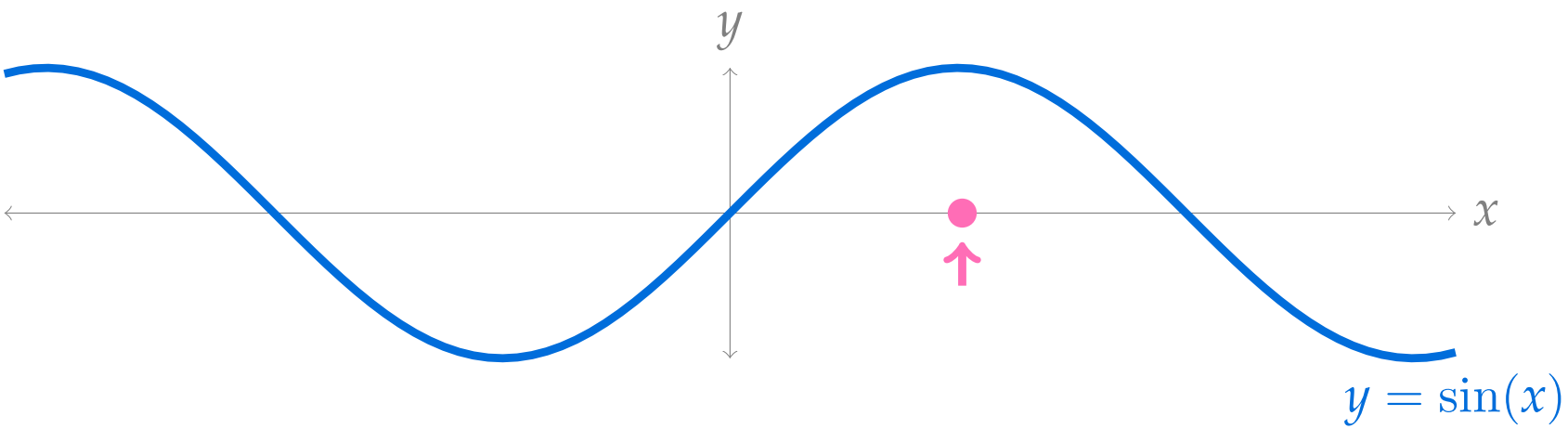
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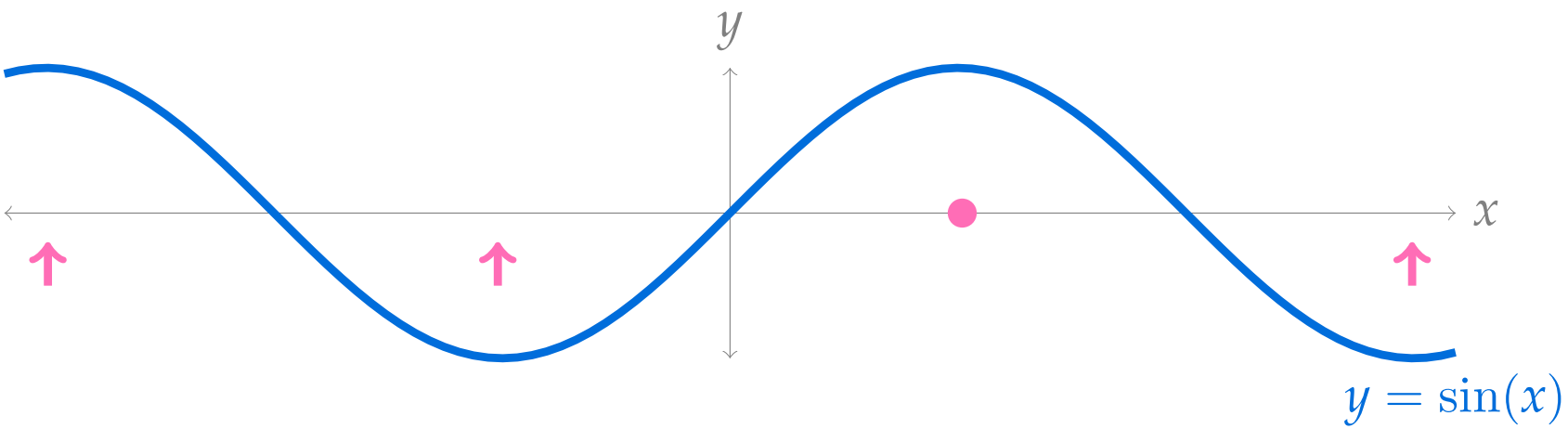
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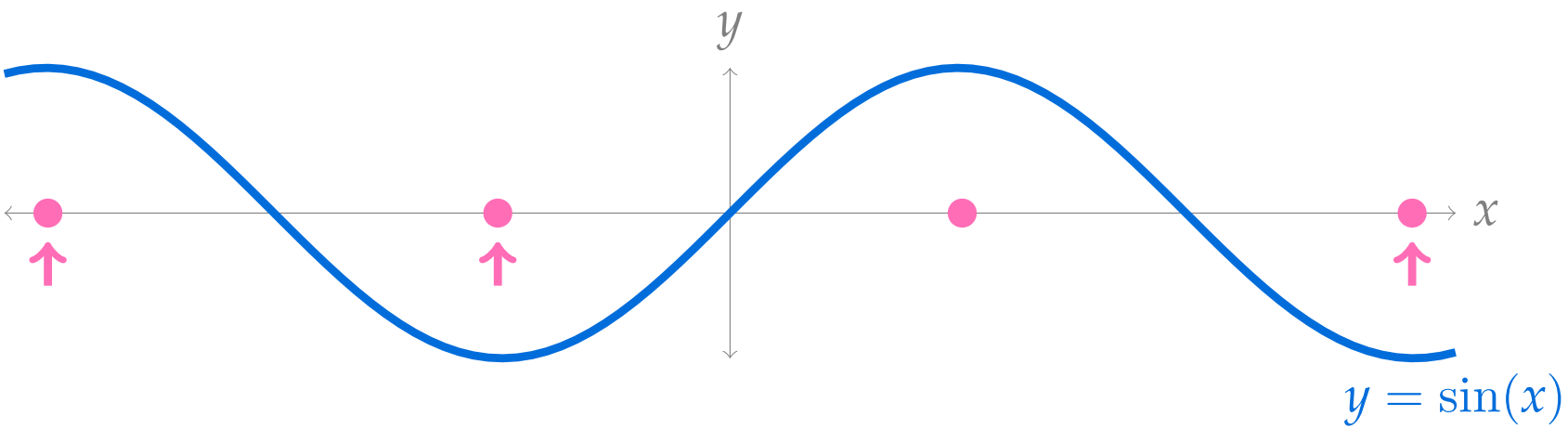
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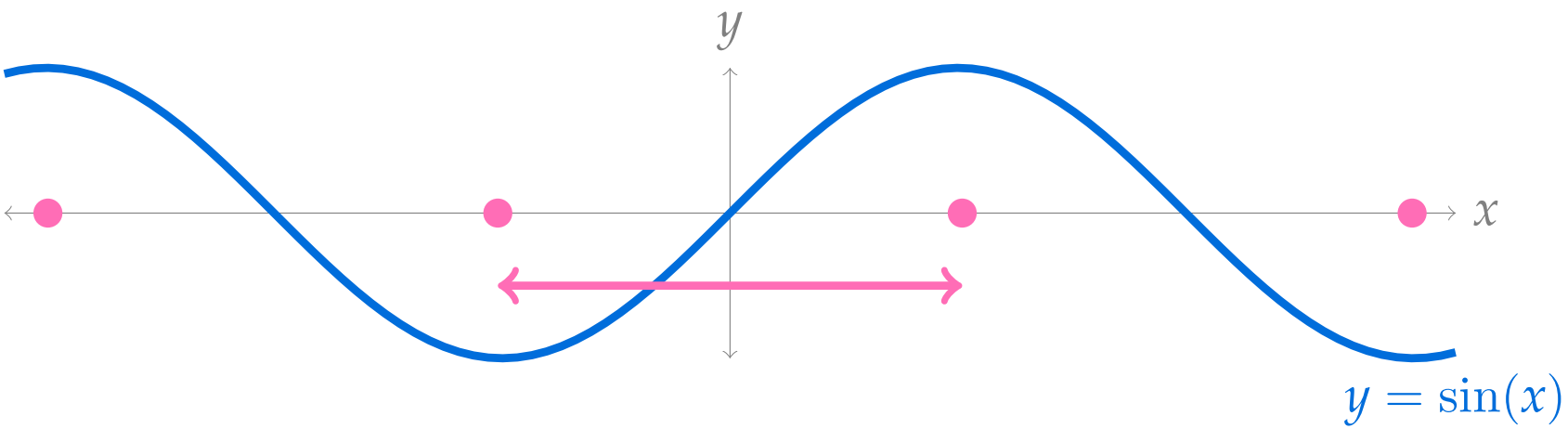
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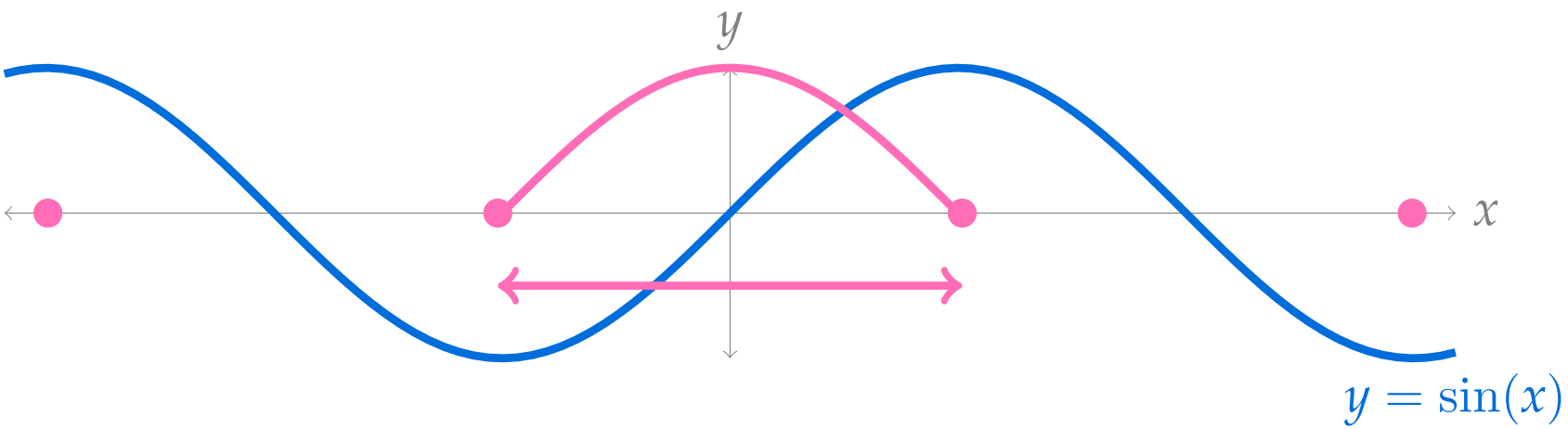
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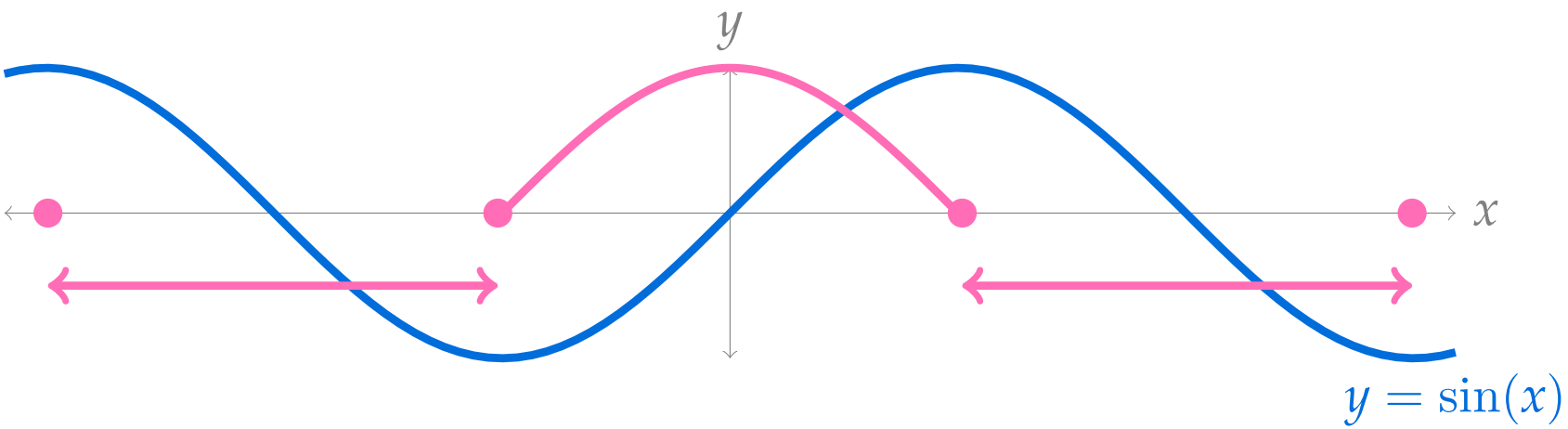
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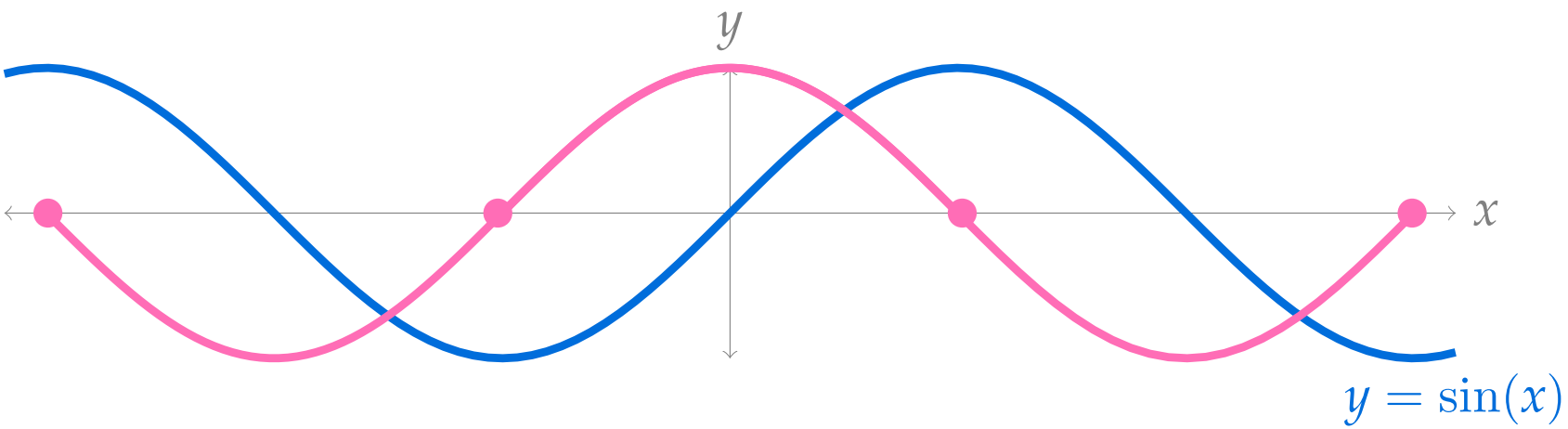
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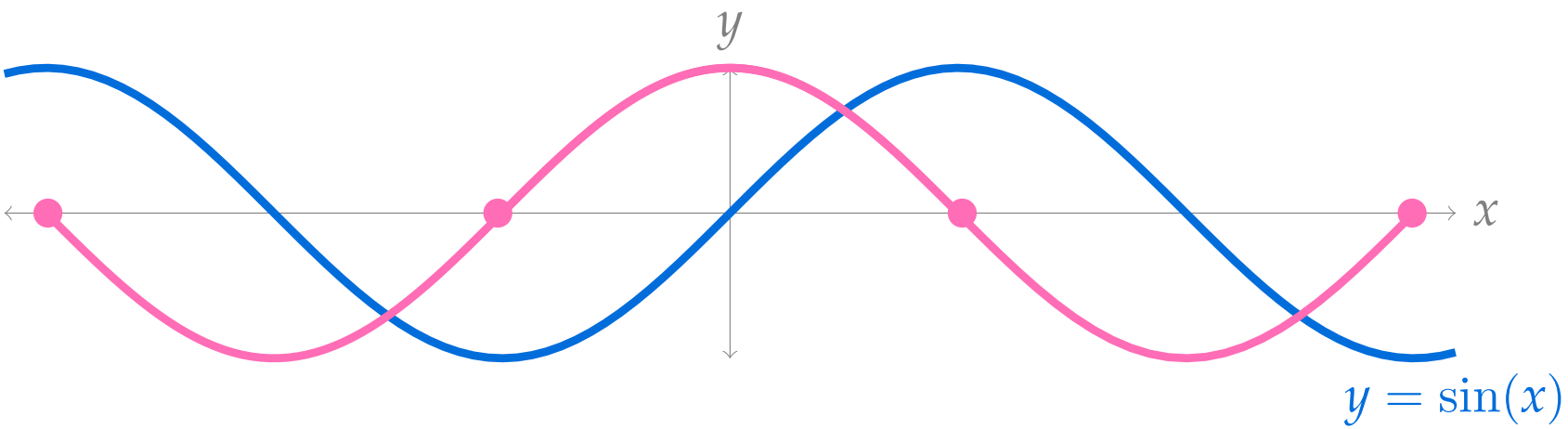
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Consider the derivative of $f(x) = \sin(x)$.

Derivative of Sine

SKIP PROOFS OF SINE AND COSINE DERIVATIVES



Consider the derivative of $f(x) = \sin(x)$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\sin(x)\} \stackrel{?}{=} \cos(x).$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\sin x\} = \lim_{h\to 0}\frac{\sin(x+h)-\sin(x)}{h}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\sin x\} = \lim_{h\to 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$= \lim_{h\to 0} \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\sin x\} = \lim_{h\to 0}\frac{\sin(x+h)-\sin(x)}{h}$$

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$$= \lim_{h\to 0}\frac{\sin(x)(\cos(h)-1)}{h} + \lim_{h\to 0}\frac{\cos(x)\sin(h)}{h}$$

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$$= \sin(x)\lim_{h\to 0}\frac{\cos(0+h)-\cos(0)}{h} + \cos(x)\lim_{h\to 0}\frac{\sin(h)}{h}$$

$$= \sin(x)\,\frac{\mathrm{d}}{\mathrm{d}x}\{\cos(x)\}\Big|_{x=0} + \cos(x)\lim_{h\to 0}\frac{\sin(h)}{h}$$

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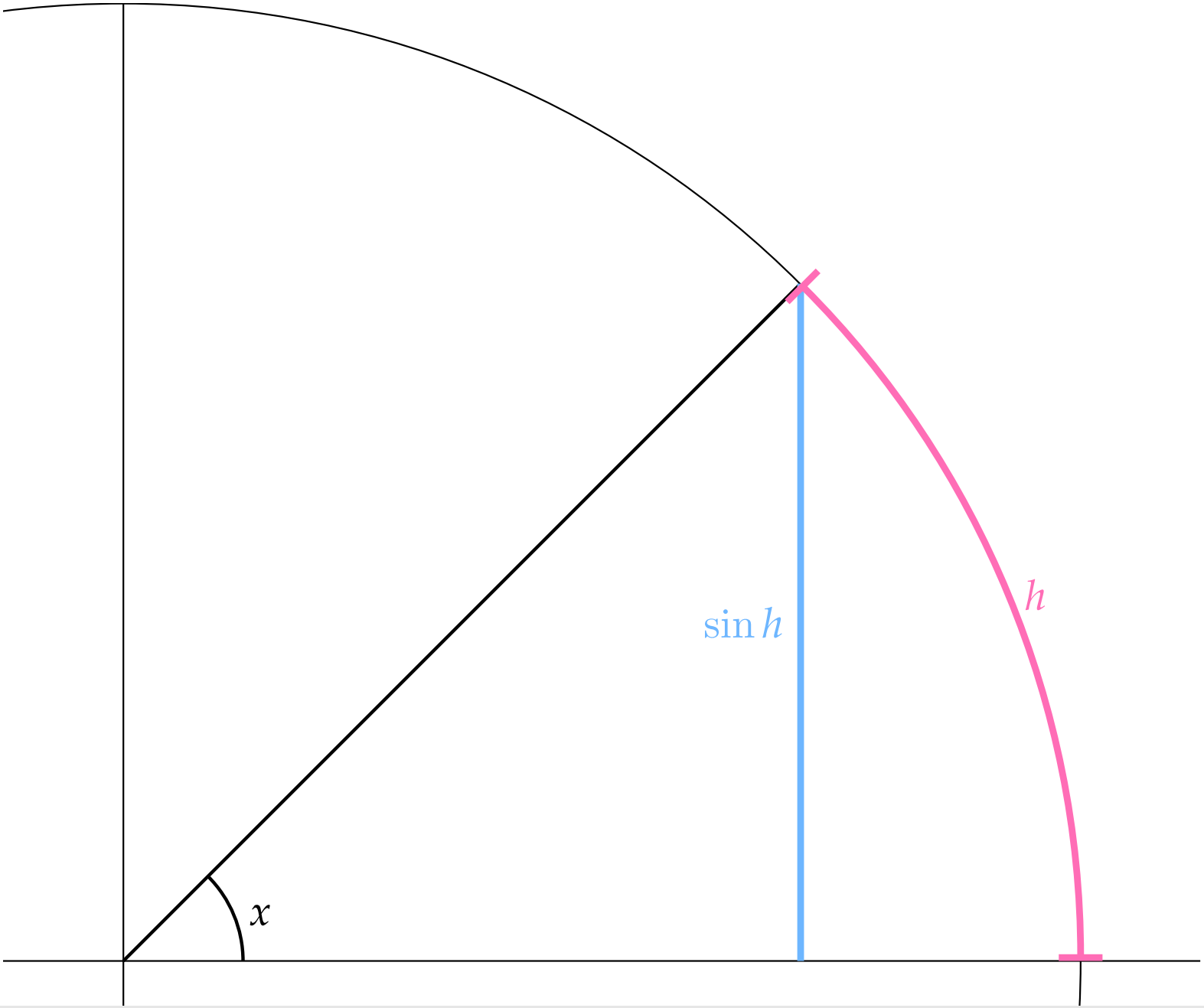
$$= \lim_{h\to 0}\frac{\sin(x)(\cos(h)-1)}{h} + \lim_{h\to 0}\frac{\cos(x)\sin(h)}{h}$$

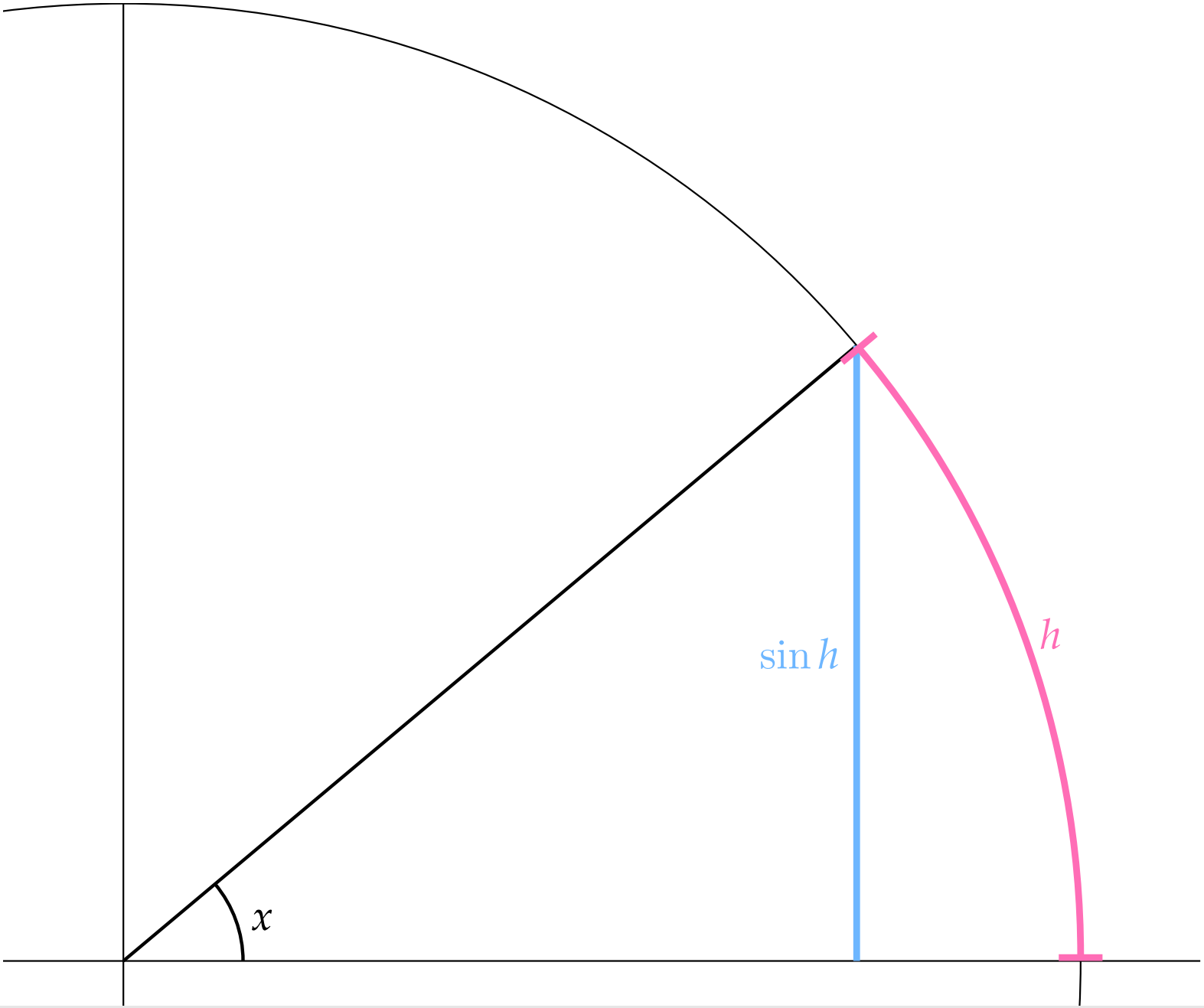
$$= \sin(x)\lim_{h\to 0}\frac{\cos(0+h)-\cos(0)}{h} + \cos(x)\lim_{h\to 0}\frac{\sin(h)}{h}$$

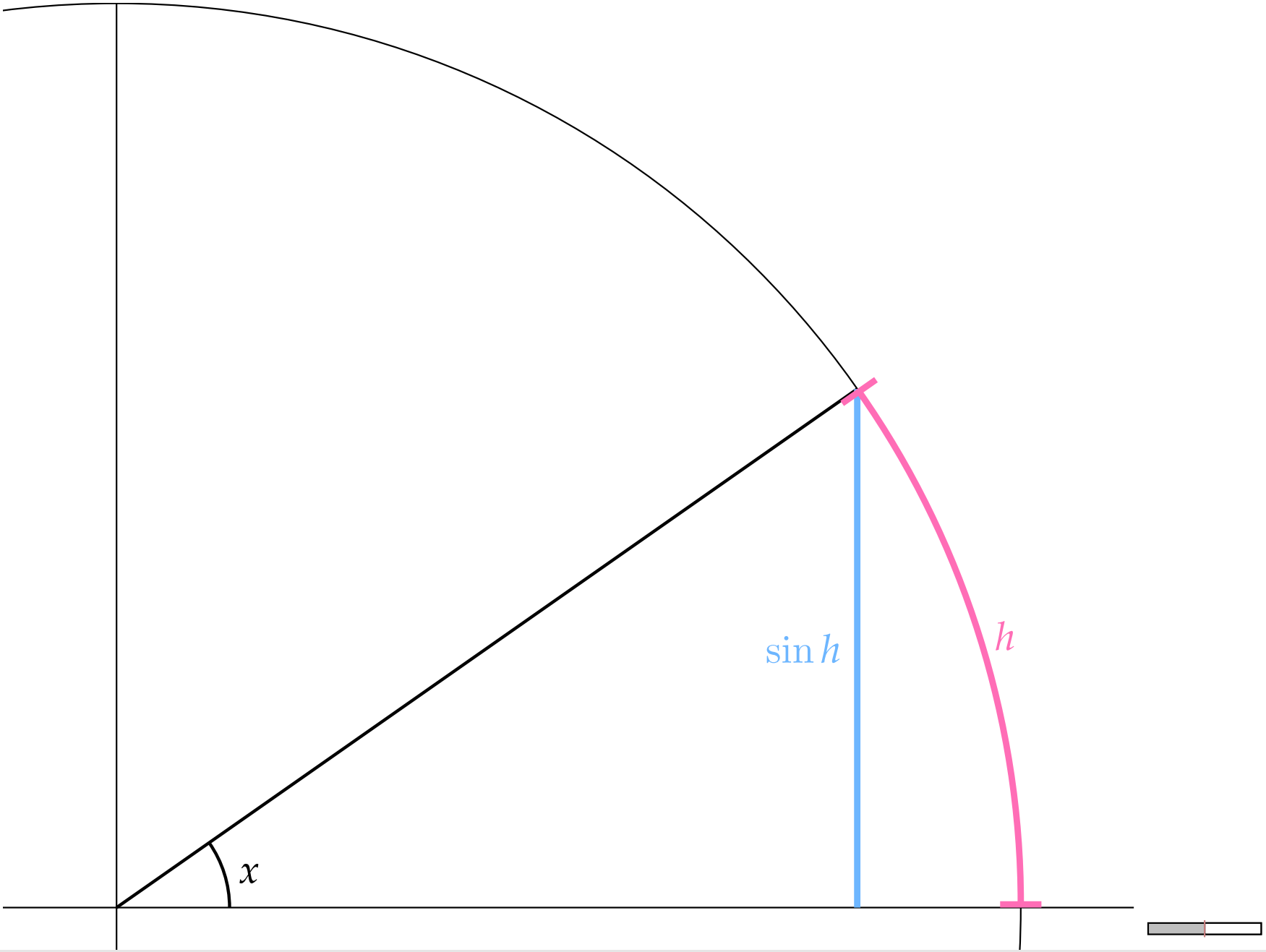
$$= \sin(x)\,\frac{\mathrm{d}}{\mathrm{d}x}\{\cos(x)\}\Big|_{x=0} + \cos(x)\lim_{h\to 0}\frac{\sin(h)}{h} \qquad = \qquad \boxed{\cos(x)\lim_{h\to 0}\frac{\sin(h)}{h}}$$

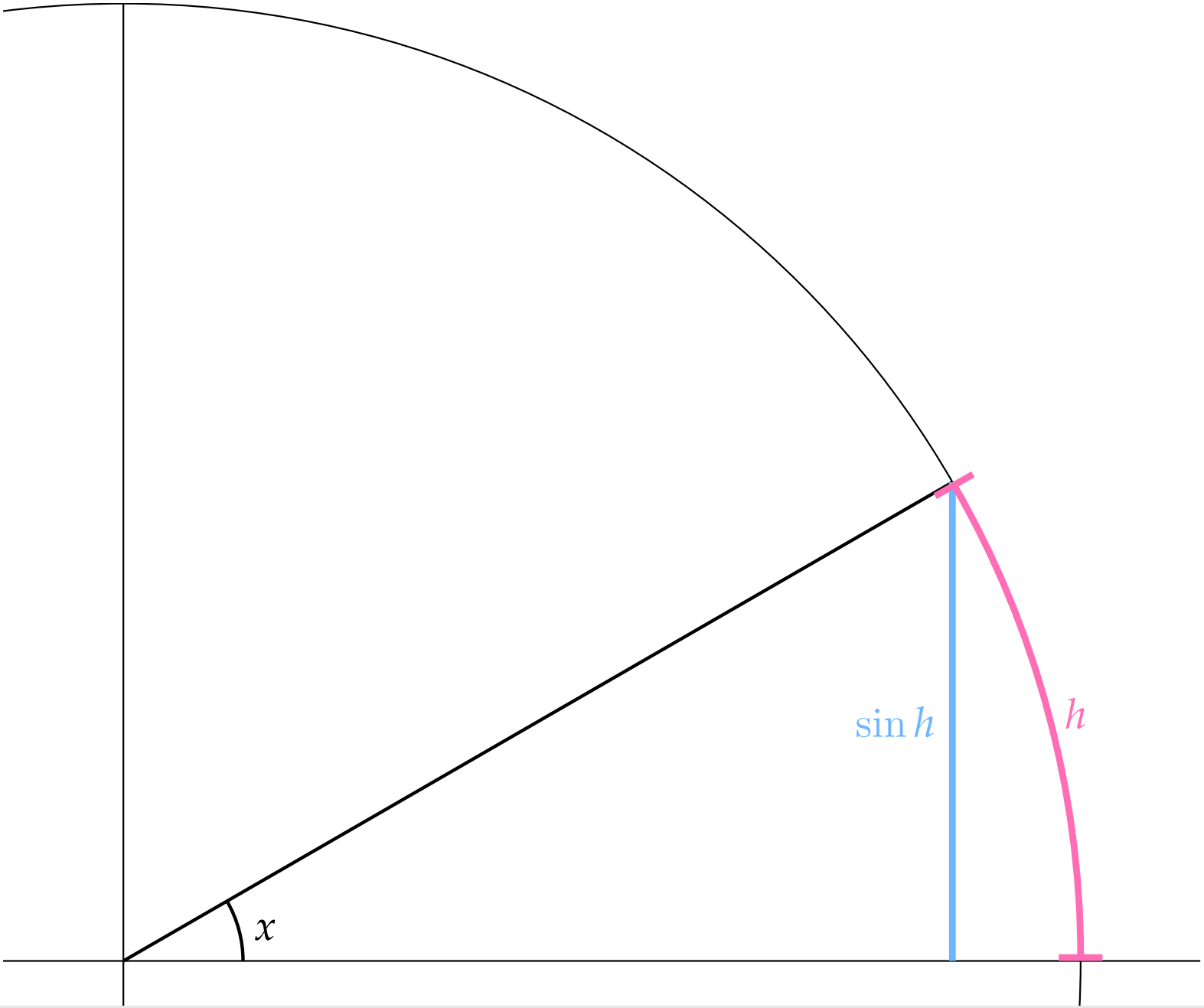
since $\cos(x)$ has a horizontal tangent, and hence has derivative zero, at $x = 0$.

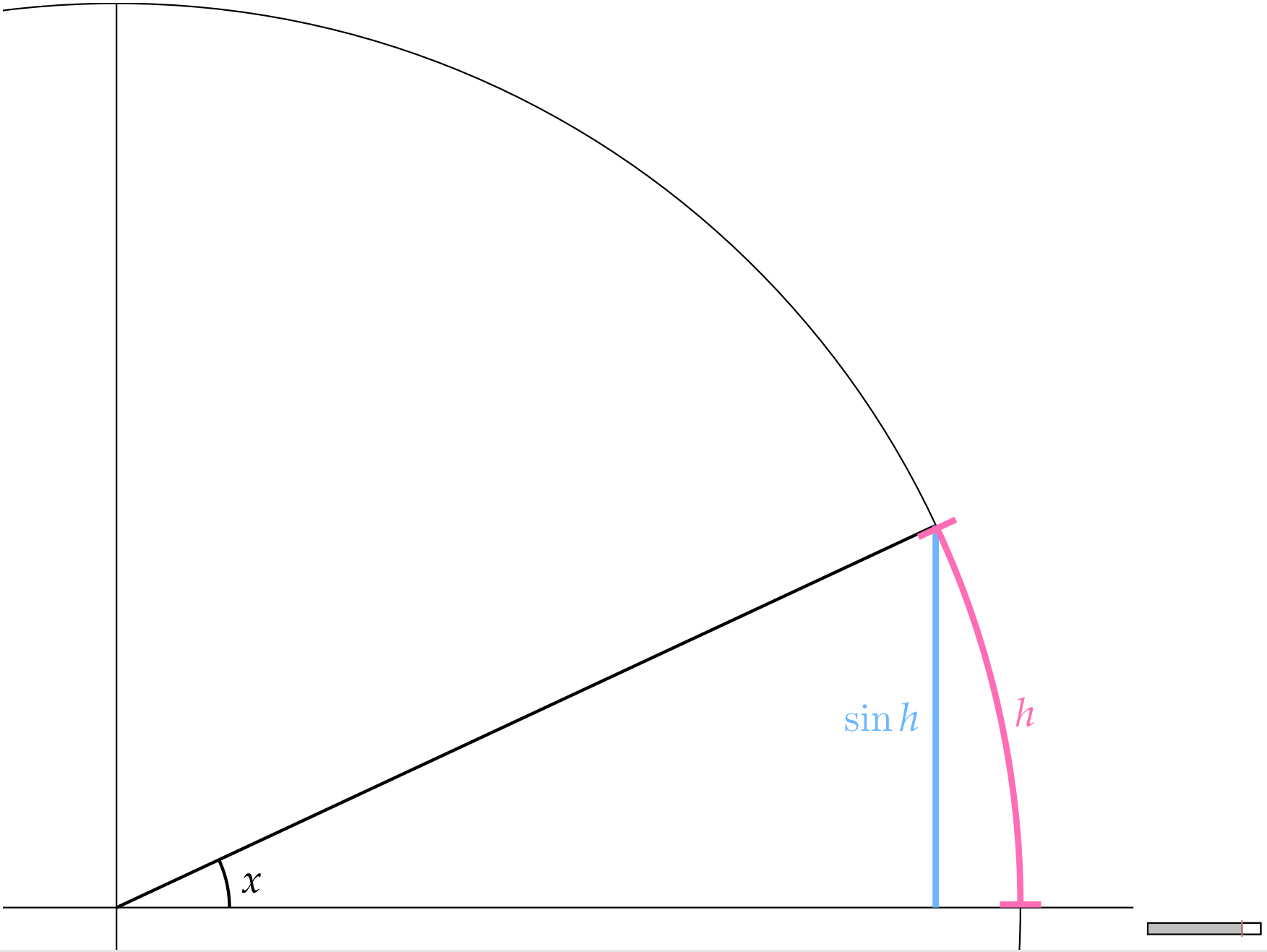
First, we investigate $\lim_{h \to 0} \frac{\sin h}{h}$ informally.

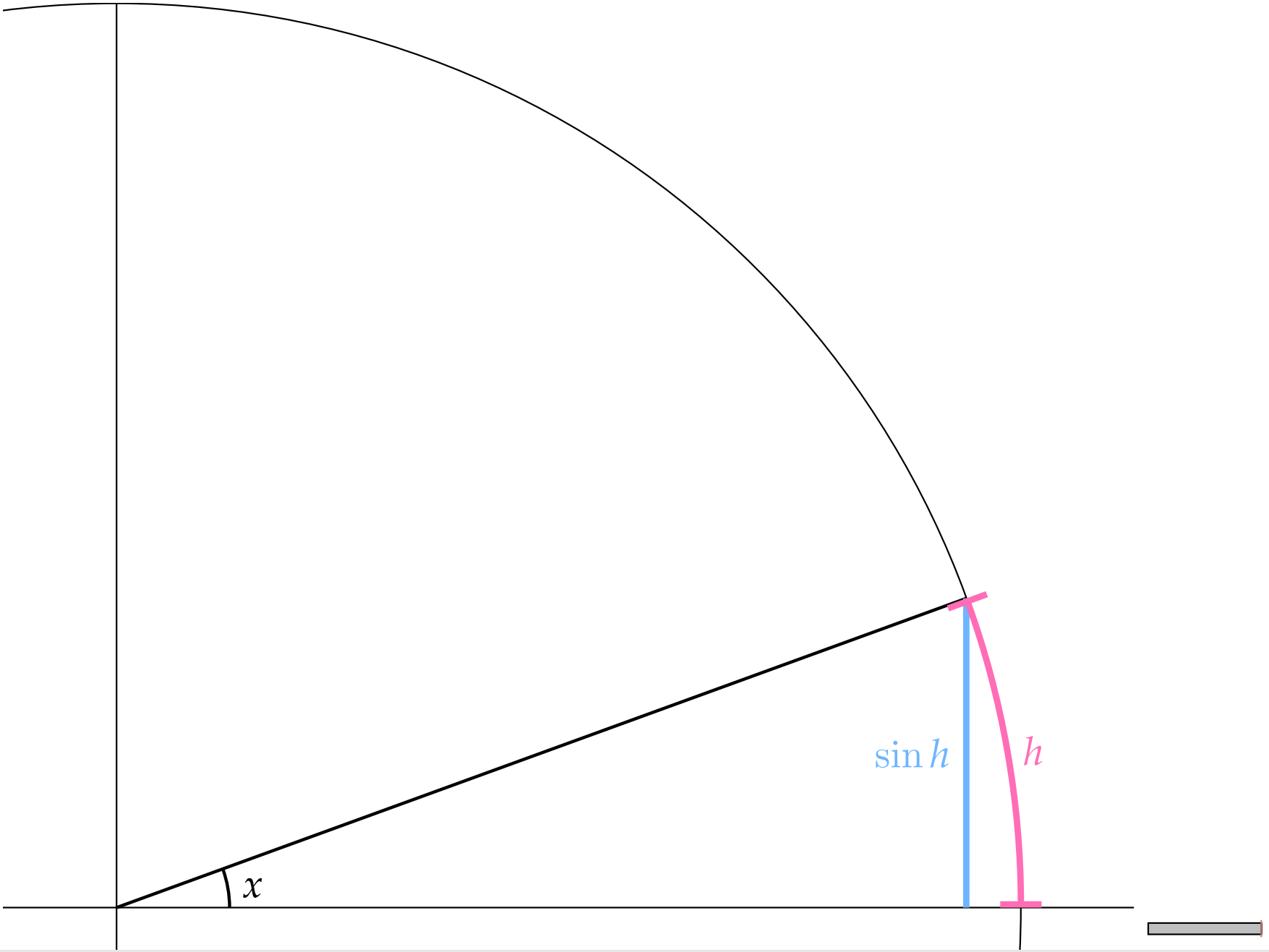

x
sin h
h

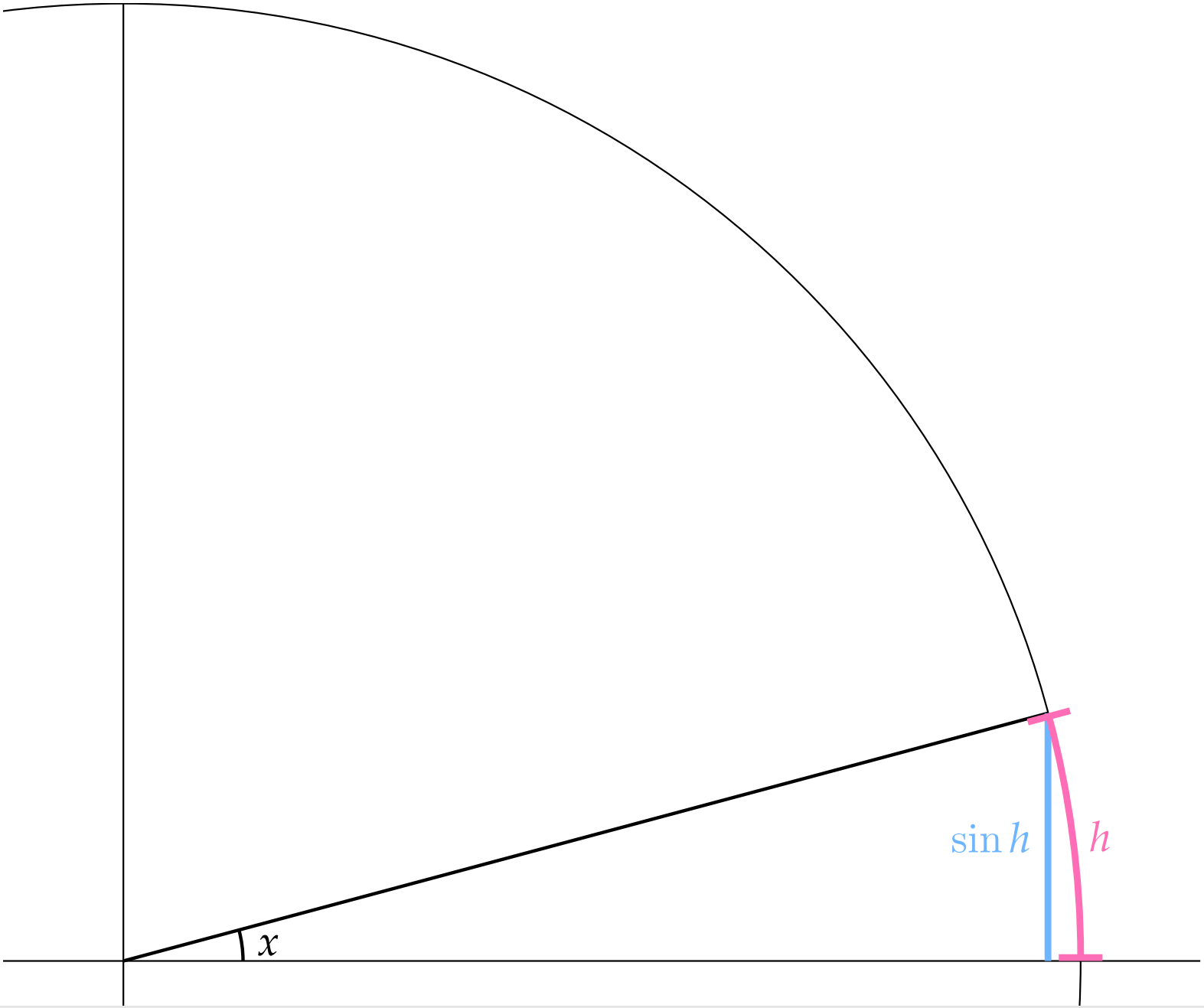

x
sin h
h

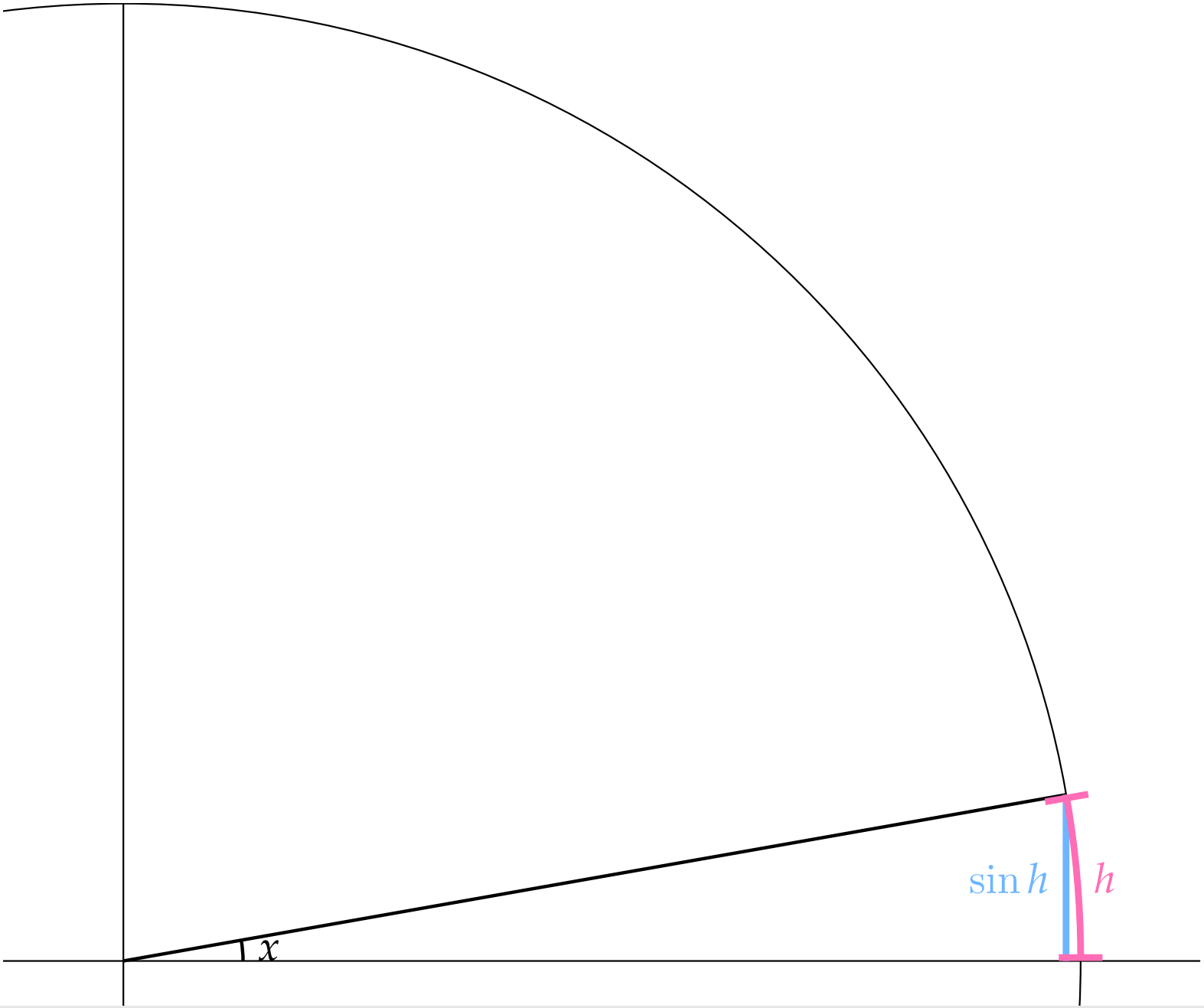

x
sin h
h


x
sin h
h


x
sin h
h


x
sin h
h

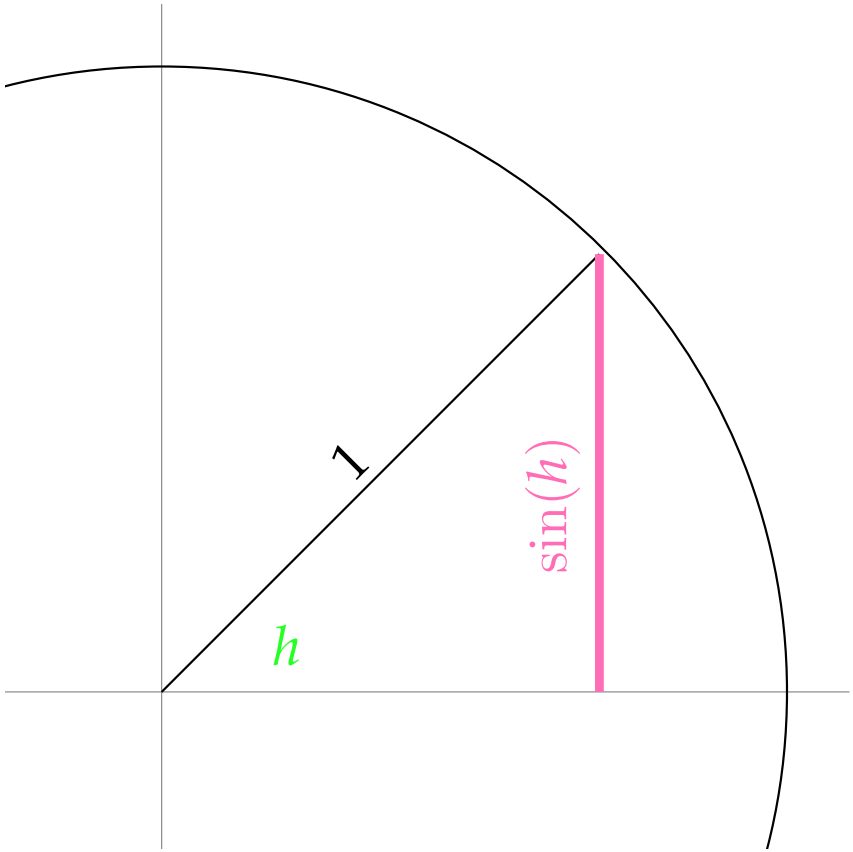

sin h
h
x

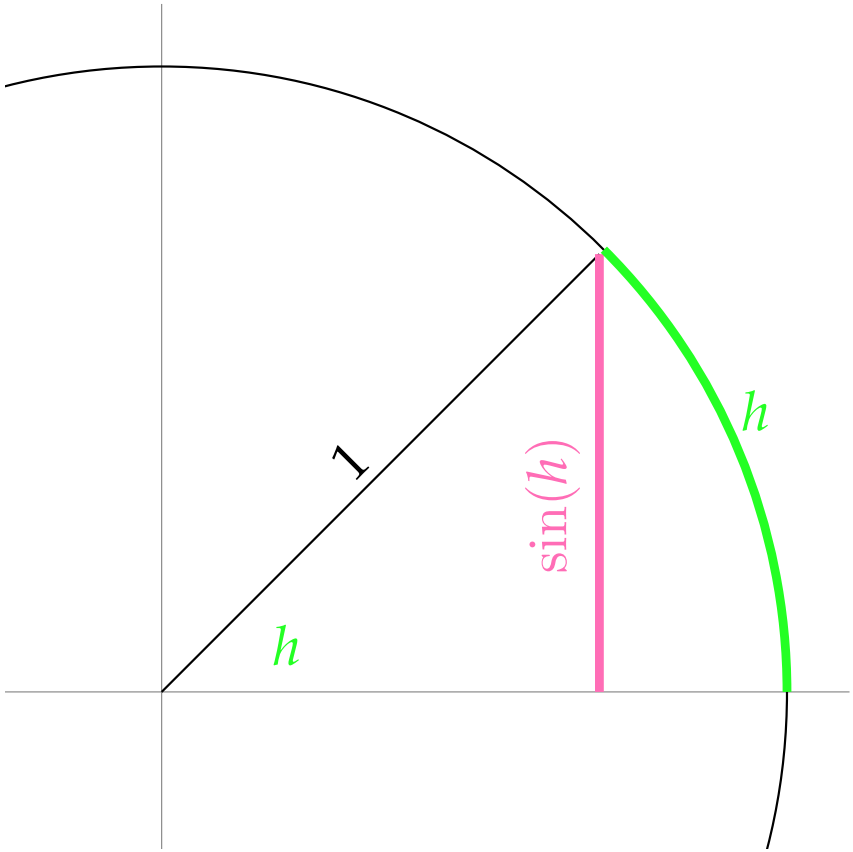

$\sin h$
h
x

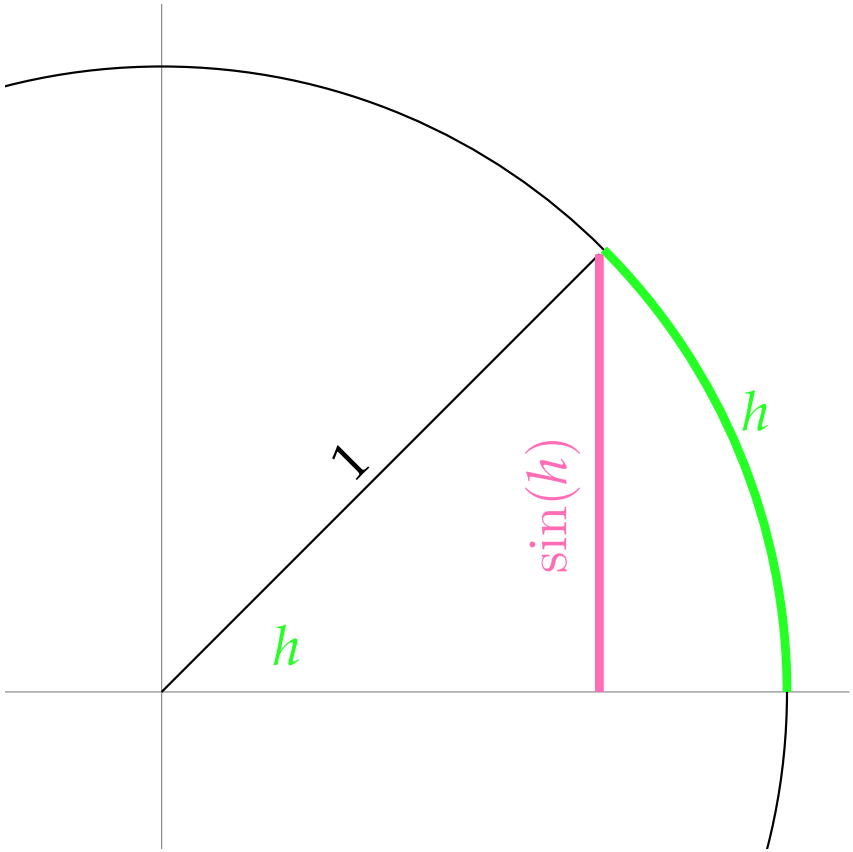
It seems $\sin h \approx h$ when $h \approx 0$, so $\lim_{h \to 0} \frac{\sin h}{h} \stackrel{?}{=} 1$.

We can prove this more formally using the Squeeze Theorem and more trigonometry. We will first prove that $\frac{\sin(h)}{h} \leq 1$ and then we will prove that $\frac{\sin(h)}{h} \geq \cos(h)$. Then we will apply the Squeeze Theorem.

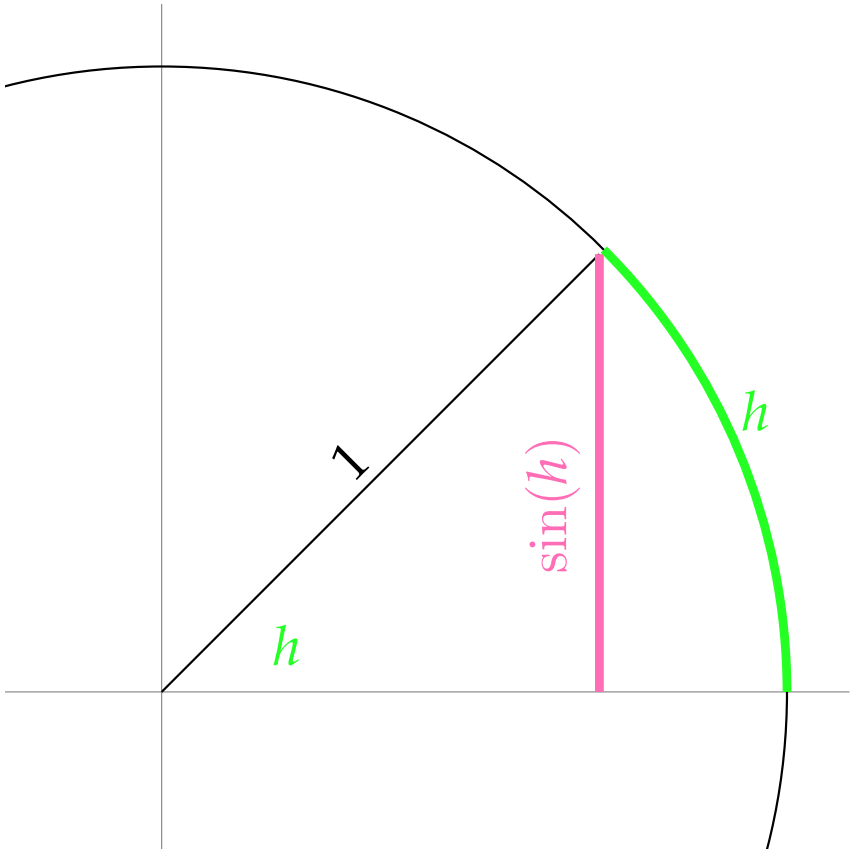
Here is the proof that $\frac{\sin(h)}{h} \leq 1$.


1
sin(h)
h


1
h
sin(h)
h

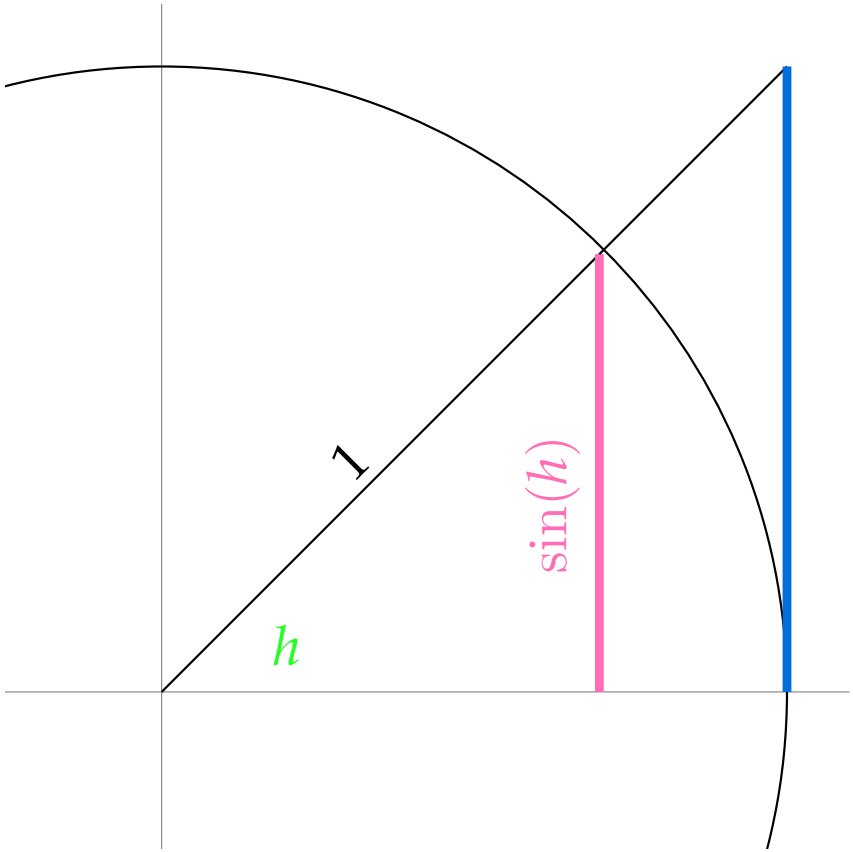


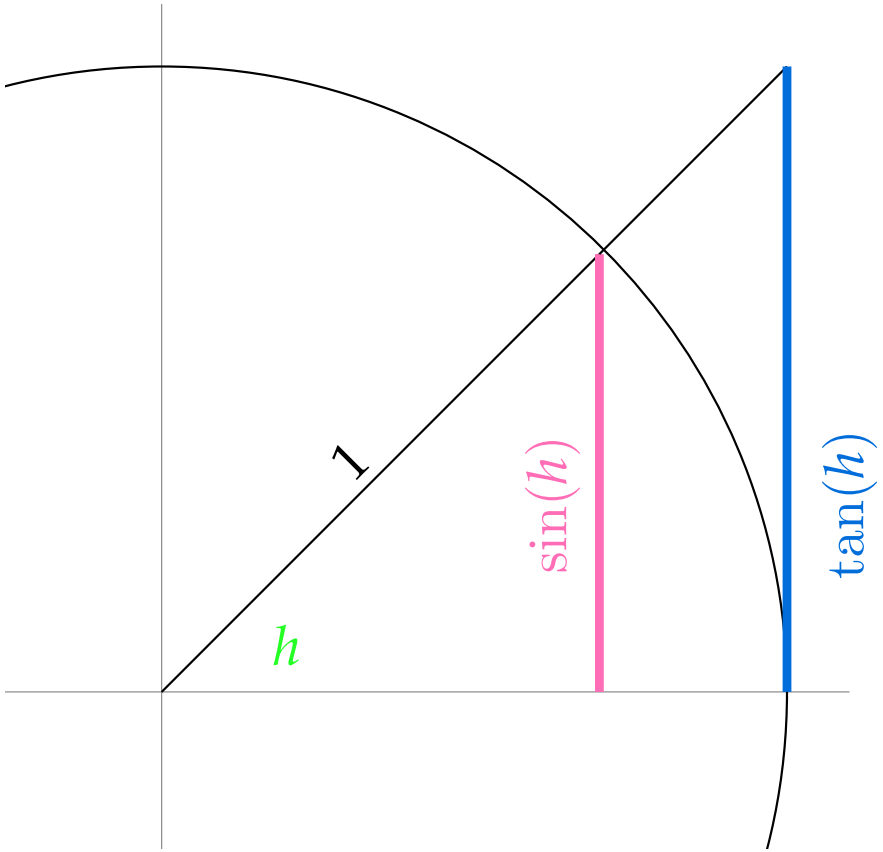
$\sin(h) \leq h$

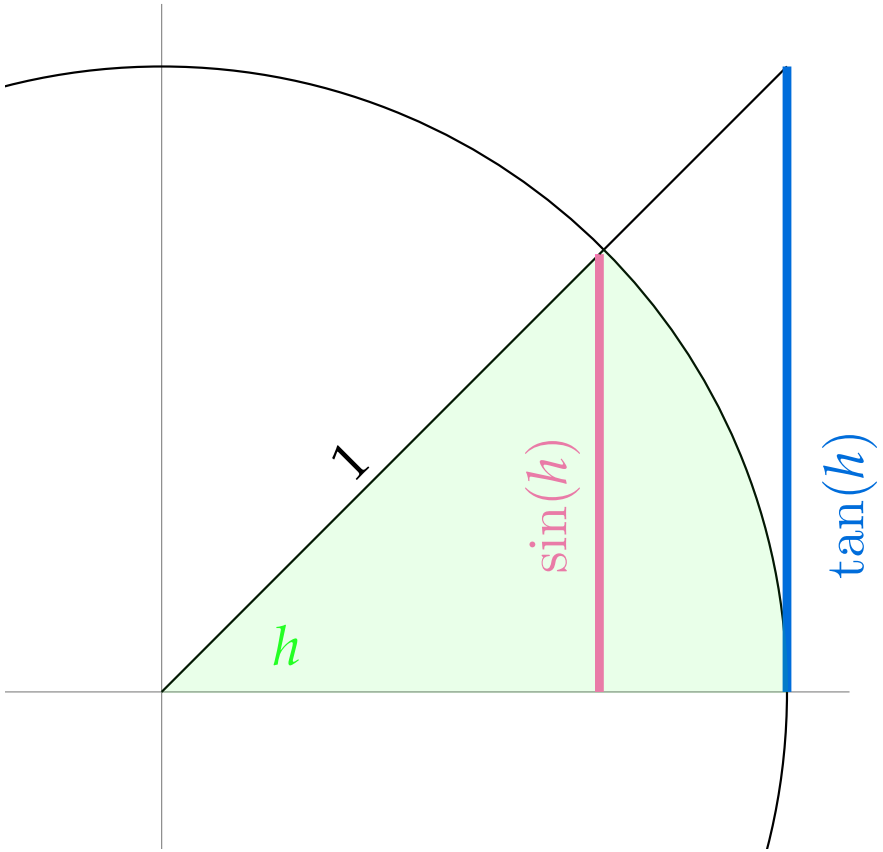


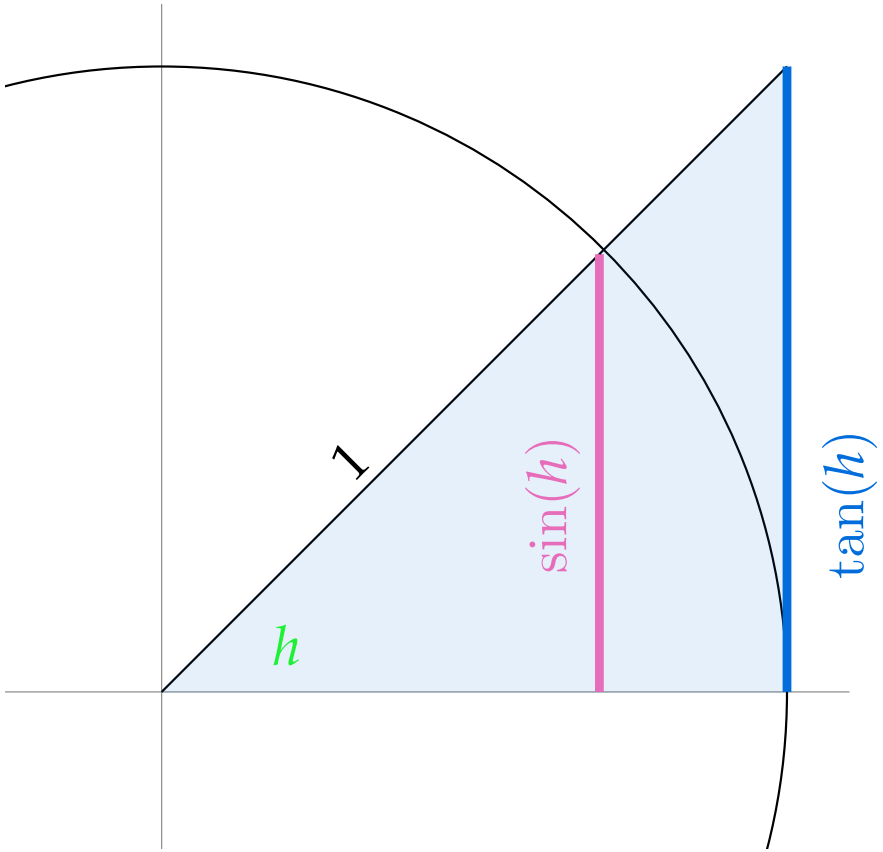
$\sin(h) \leq h$ so $\boxed{\dfrac{\sin(h)}{h} \leq 1}$

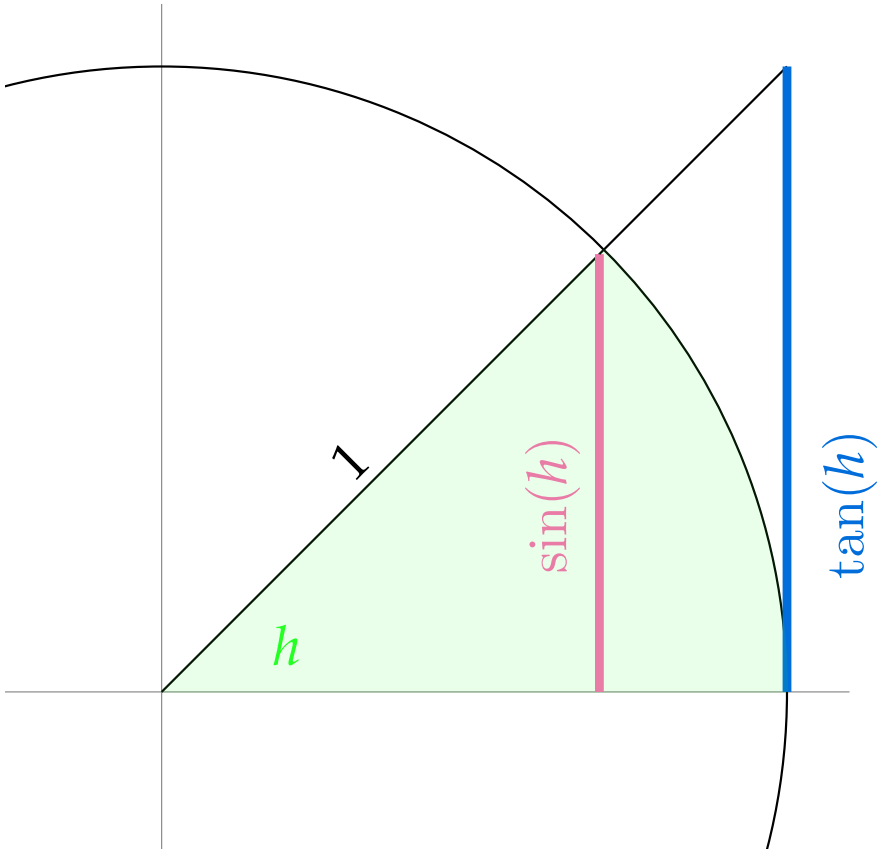
Now for the proof that $\dfrac{\sin(h)}{h} \geq \cos(h)$.


1
sin(h)
h

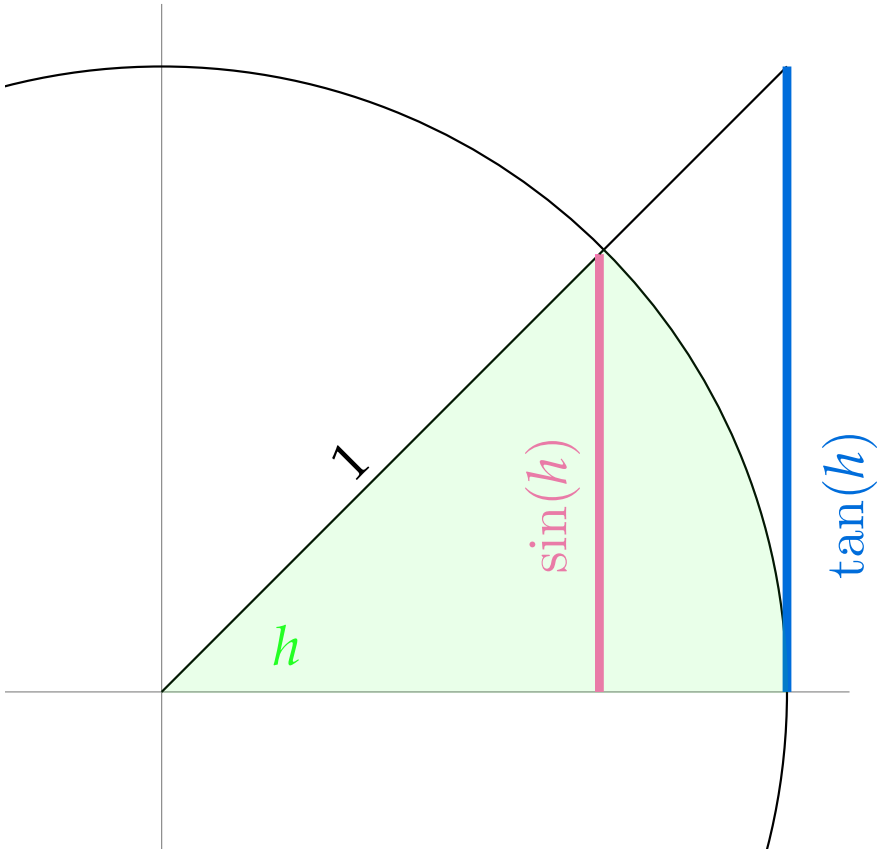

1
h
sin(h)
tan(h)


1
sin(h)
tan(h)
h

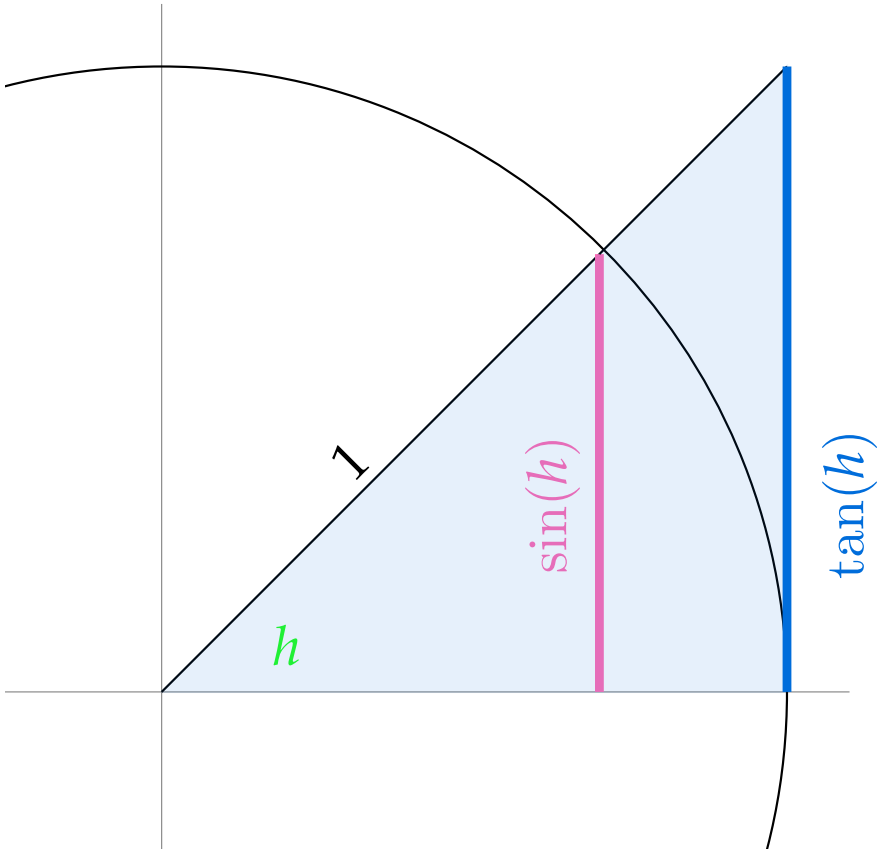

1
sin(h)
tan(h)
h



Green area:

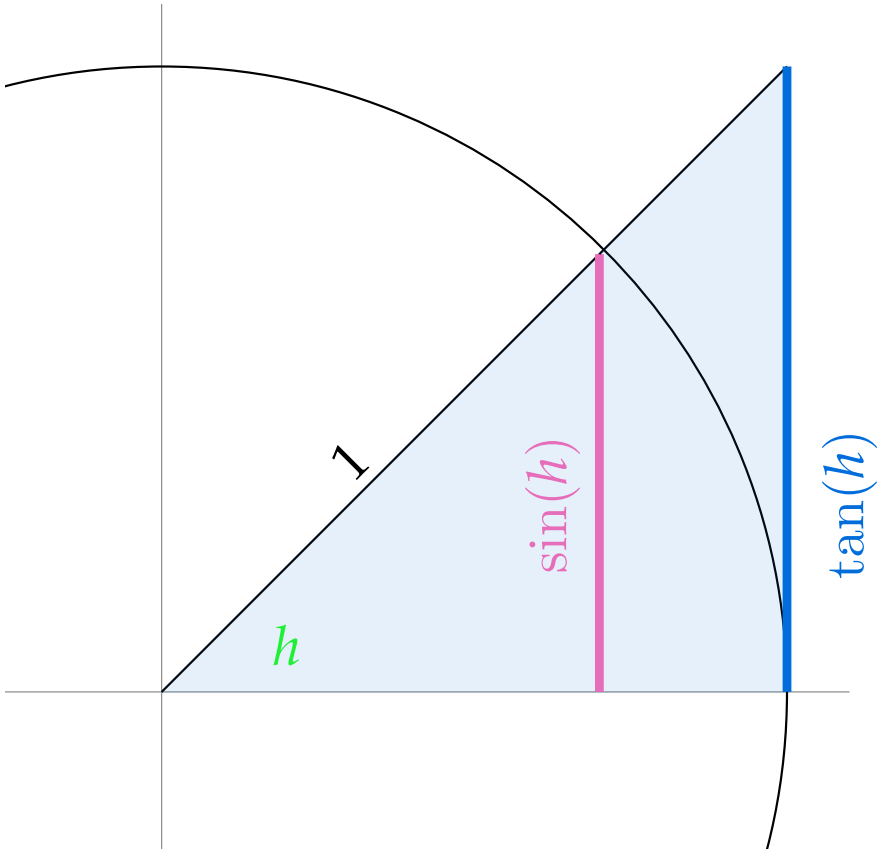


Green area: $\frac{h}{2}$



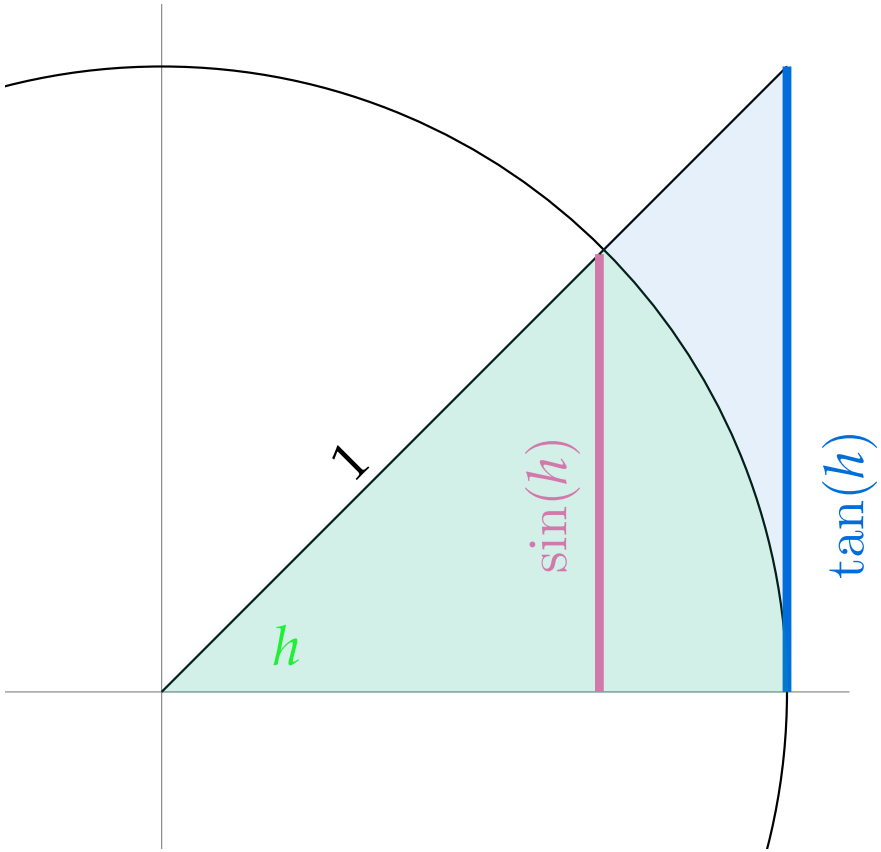
Green area: $\frac{h}{2}$

Blue area:

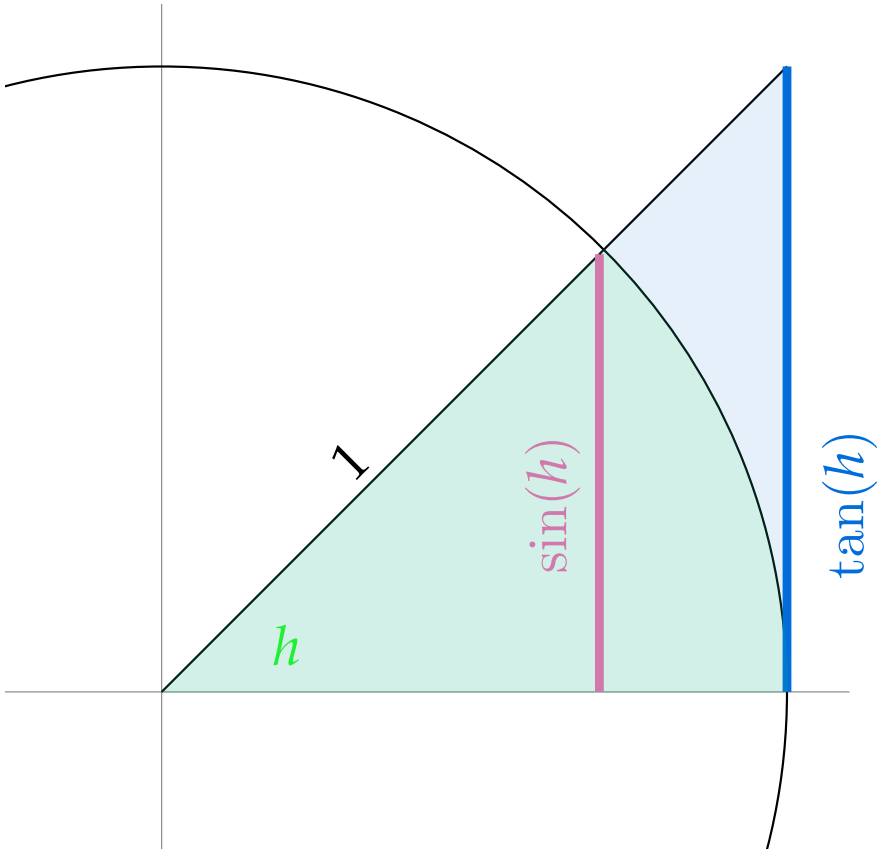


Green area: $\frac{h}{2}$

Blue area: $\frac{\tan h}{2}$



Green area: $\frac{h}{2}$ $\frac{h}{2} \leq \frac{\tan(h)}{2}$ Blue area: $\frac{\tan h}{2}$



Green area: $\frac{h}{2}$

$$\frac{h}{2} \leq \frac{\tan(h)}{2}$$

Blue area: $\frac{\tan h}{2}$

$$\boxed{\cos(h) \leq \frac{\sin(h)}{h}}$$

We are now ready for the Squeeze Theorem. We have

$$\cos h \quad \leq \quad \frac{\sin h}{h} \quad \leq \quad 1$$

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$$\begin{array}{ccccc} \cos h & \leq & \dfrac{\sin h}{h} & \leq & 1 \\ \lim\limits_{h\to 0} \cos h = 1 & & & & \lim\limits_{h\to 0} 1 = 1 \end{array}$$

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By the Squeeze Theorem,

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By the Squeeze Theorem,

$$\lim_{h\to 0} \frac{\sin h}{h} = 1$$

Derivatives of Sine and Cosine

From before,

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\sin(x)\} = \cos(x) \cdot \lim_{h \to 0} \frac{\sin(h)}{h} =$$

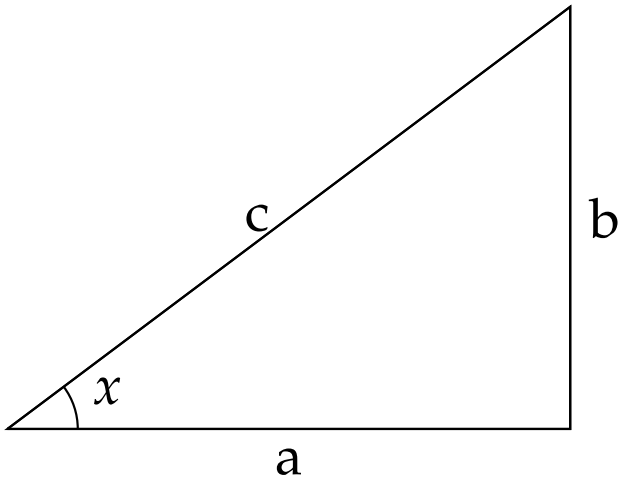
Derivatives of Sine and Cosine

From before,

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\sin(x)\} = \cos(x) \cdot \lim_{h \to 0} \frac{\sin(h)}{h} = \cos(x)$$

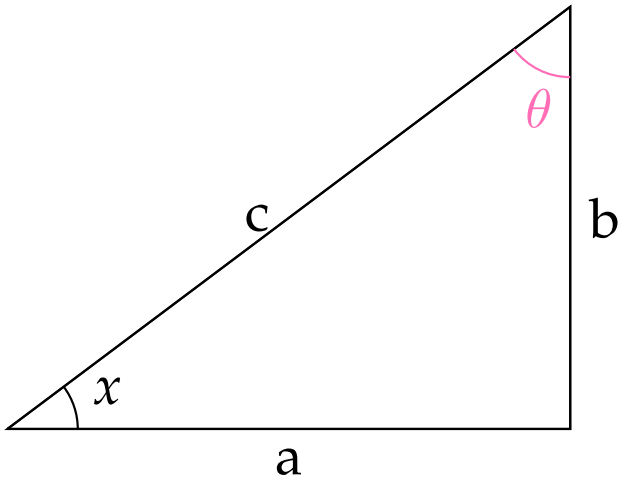
Derivative of Cosine

Now for the derivative of cos. We already know the derivative of sin, and it is easy to convert between sin and cos using trig identities.



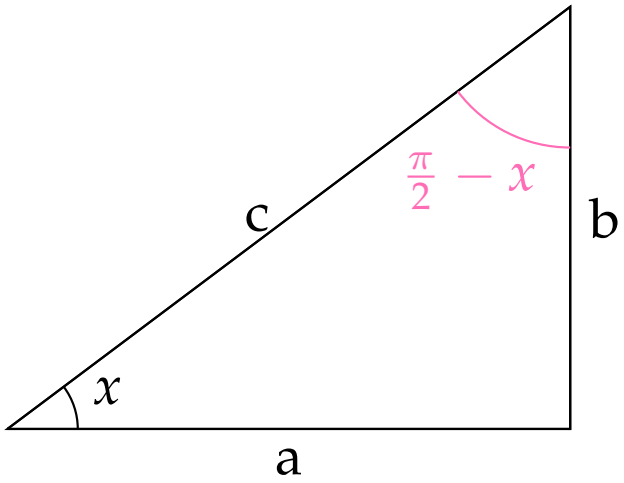
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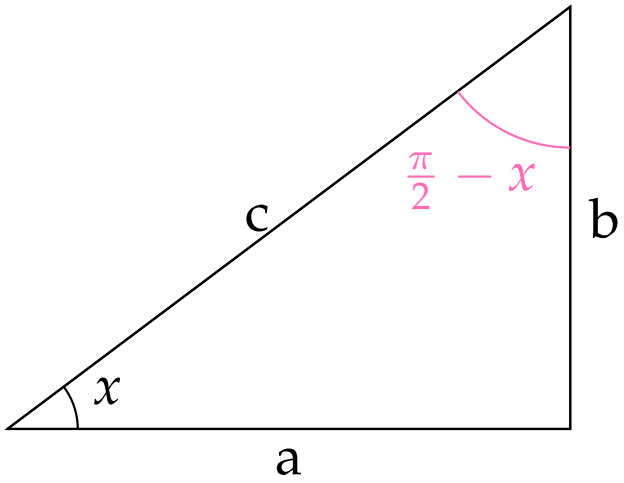
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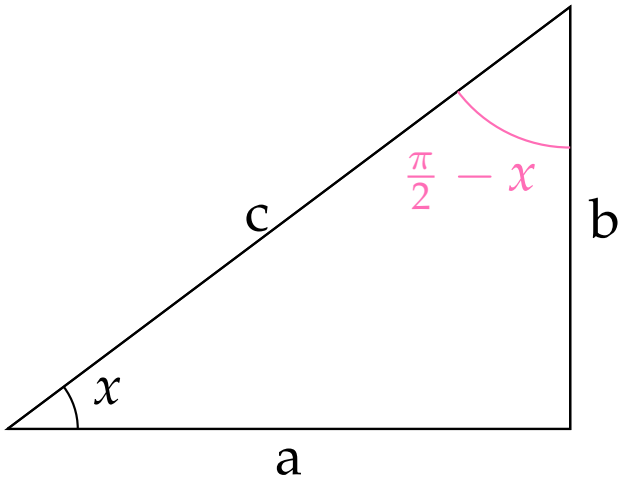
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$$\sin x = \frac{b}{c} = \cos\left(\frac{\pi}{2} - x\right)$$
$$\cos x = \frac{a}{c} = \sin\left(\frac{\pi}{2} - x\right)$$

Derivative of Cosine

Now for the derivative of cos. We already know the derivative of sin, and it is easy to convert between sin and cos using trig identities.



$$\sin x = \frac{b}{c} = \cos\left(\frac{\pi}{2} - x\right)$$
$$\cos x = \frac{a}{c} = \sin\left(\frac{\pi}{2} - x\right)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\cos(x)\right] = \frac{\mathrm{d}}{\mathrm{d}x}\left[\sin\left(\tfrac{\pi}{2} - x\right)\right] = -\frac{\mathrm{d}}{\mathrm{d}x}\left[\sin\left(x - \tfrac{\pi}{2}\right)\right] = -\cos\left(x - \tfrac{\pi}{2}\right) = -\sin x$$

since $\sin(-\theta) = -\sin(\theta)$ and $\cos(-\theta) = \cos(\theta)$.

When we use radians:

Derivatives of Trig Functions

$\frac{d}{dx}\{\sin(x)\} = \cos(x)$

$\frac{d}{dx}\{\cos(x)\} = -\sin(x)$

$\frac{d}{dx}\{\tan(x)\} =$

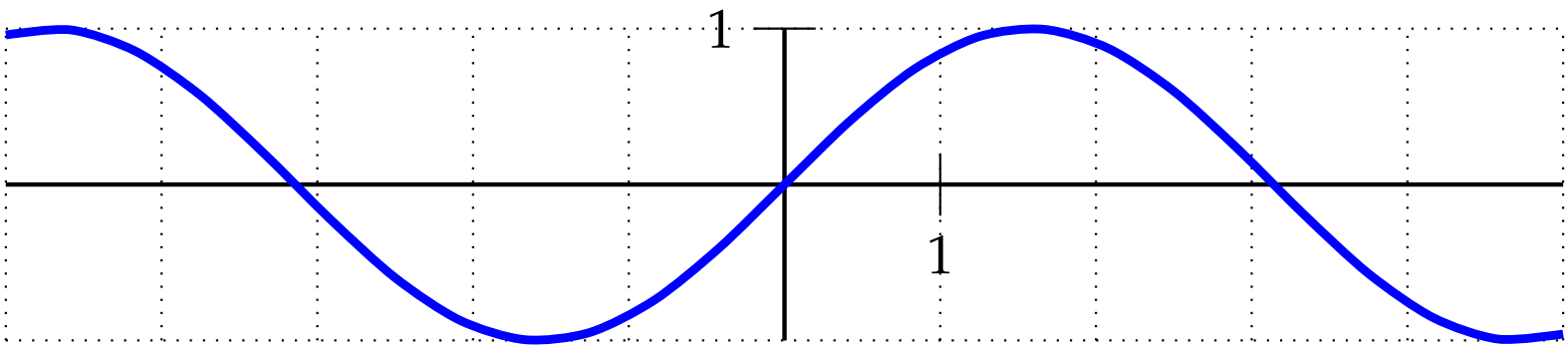
$\frac{d}{dx}\{\sec(x)\} =$

$\frac{d}{dx}\{\csc(x)\} =$

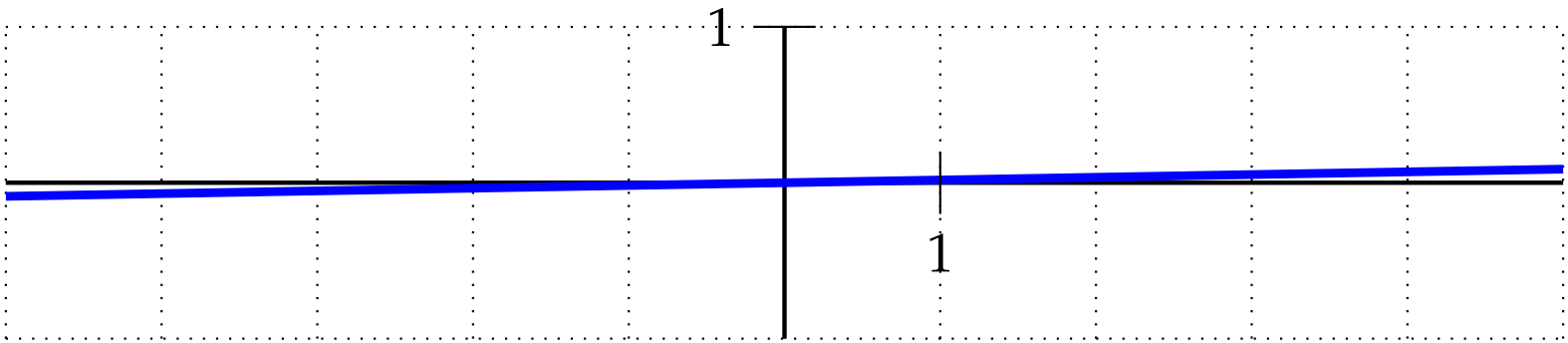
$\frac{d}{dx}\{\cot(x)\} =$

Honourable Mention

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$



$y = \sin x$, radians



$y = \sin x$, degrees

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

ans

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}[\tan(x)] &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\sin(x)}{\cos(x)}\right] \\
&= \frac{\cos(x)\cos(x) - \sin(x)[-\sin(x)]}{\cos^2(x)} \\
&= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} \\
&= \frac{1}{\cos^2(x)} = \sec^2(x)
\end{aligned}$$

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\sec(x) = \frac{1}{\cos(x)}$$

ans

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\sec(x) = \frac{1}{\cos(x)}$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}[\sec(x)] &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{\cos(x)}\right] \\
&= \frac{\cos(x)(0) - (1)(-\sin(x))}{\cos^2(x)} \\
&= \frac{\sin(x)}{\cos^2(x)} \\
&= \frac{1}{\cos(x)}\frac{\sin(x)}{\cos(x)} \\
&= \sec(x)\tan(x)
\end{aligned}$$

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\csc(x) = \frac{1}{\sin(x)}$$

ans

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\csc(x) = \frac{1}{\sin(x)}$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}[\csc(x)] &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{\sin(x)}\right] \\
&= \frac{\sin(x)(0) - (1)\cos(x)}{\sin^2(x)} \\
&= \frac{-\cos(x)}{\sin^2(x)} \\
&= \frac{-1}{\sin(x)}\frac{\cos(x)}{\sin(x)} \\
&= -\csc(x)\cot(x)
\end{aligned}$$

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

ans

Other Trig Functions

SKIP PROOFS OF OTHER TRIG DERIVATIVES

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}[\cot(x)] &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\cos(x)}{\sin(x)}\right] \\
&= \frac{\sin(x)(-\sin(x)) - \cos(x)\cos(x)}{\sin^2(x)} \\
&= \frac{-1}{\sin^2(x)} \\
&= -\csc^2(x)
\end{aligned}$$

Memorize

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\sin(x)\} = \cos(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\{\sec(x)\} = \sec(x)\tan(x)$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\{\cos(x)\} = -\sin(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\{\csc(x)\} = -\csc(x)\cot(x)$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\{\tan(x)\} = \sec^2(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\{\cot(x)\} = -\csc^2(x)$$

$$\lim_{x\to 0}\frac{\sin x}{x} = 1$$

NOW YOU



1. Let $f(x) = \dfrac{x\tan(x^2+7)}{15e^x}$. Use the **definition of the derivative** to find $f'(0)$.

2. Differentiate $\left(e^x + \cot x\right)\left(5x^6 - \csc x\right)$.

3. Let $h(x) = \begin{cases} \frac{\sin x}{x} & , \quad x < 0 \\ \frac{ax+b}{\cos x} & , \quad x \geq 0 \end{cases}$
   Which values of a and b make $h(x)$ continuous at $x = 0$?

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Which values of a and b make $h(x)$ continuous at $x = 0$?

Practice and Review

$$f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right) & , \quad x \neq 0 \\ 0 & , \quad x = 0 \end{cases}$$

Is $f(x)$ differentiable at $x = 0$?

$$g(x) = \begin{cases} e^{\frac{\sin x}{x}} & , \quad x < 0 \\ (x-a)^2 & , \quad x \geq 0 \end{cases}$$

What value(s) of a makes $g(x)$ continuous at $x = 0$?

We don't have rules for differentiating $f(x)$ at $x = 0$, so we have to fall back on the definition of the derivative.

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{h^2 \cos\left(\frac{1}{h}\right) - 0}{h}$$
$$= \lim_{h\to 0} h \cos\left(\frac{1}{h}\right) = 0$$

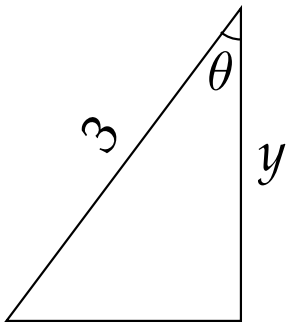
Since the limit exists, $f(x)$ is differentiable at 0.
By the definition of continuity, $g(x)$ is continuous at $x = 0$ if

$$\lim_{x\to 0} g(x) = g(0)$$

- $g(0) = (0 - a)^2 = a^2$
- $\lim\limits_{x\to 0^-} g(x) = \lim\limits_{x\to 0^-} e^{\frac{\sin x}{x}} = e^1 = e$
- $\lim\limits_{x\to 0^+} g(x) = \lim\limits_{x\to 0^+} (x - a)^2 = a^2$

In order for $g(x)$ to be continuous, we need $a^2 = e$. That is, $a = \sqrt{e}$ or $a = -\sqrt{e}$.

A ladder 3 metres long rests against a vertical wall. Let θ be the angle between the top of the ladder and the wall, measured in radians, and let y be the height of the top of the ladder. If the ladder slides away from the wall, how fast does y change with respect to θ?
When is the top of the ladder sinking the fastest? The slowest?



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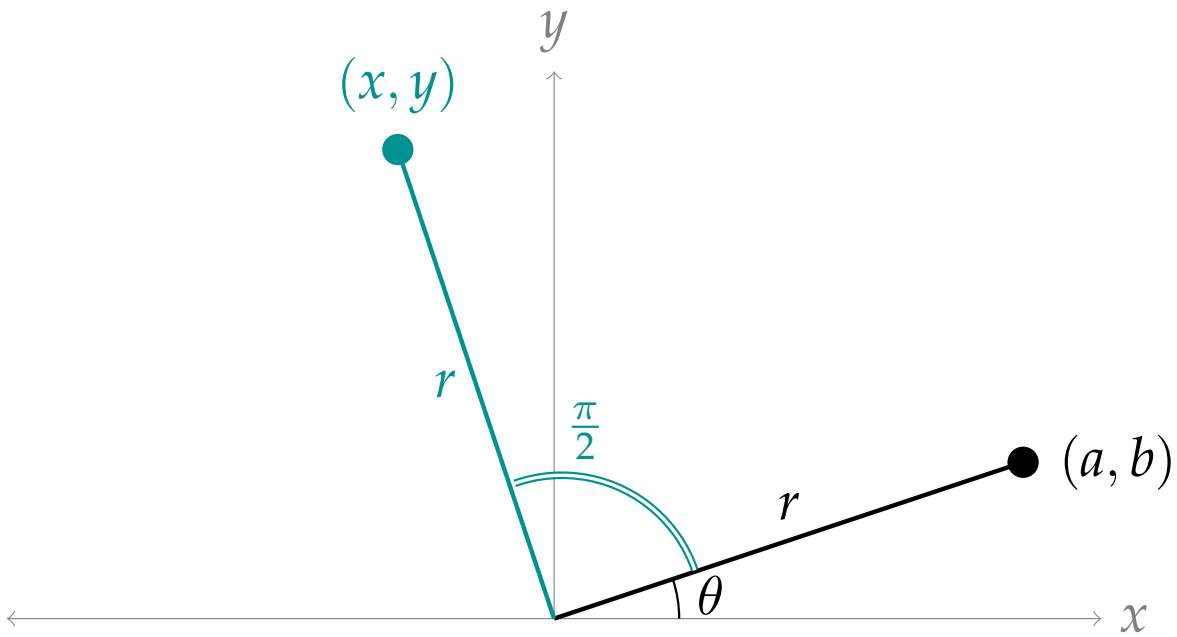
We want to find how fast y is changing with respect to θ, so we want $\frac{dy}{d\theta}$, or $y'(\theta)$. To calculate that, we need to find y as a function of θ. Note that the ladder forms a right triangle with the wall, and y is the side adjacent to θ, while 3 is the hypotenuse. So, $\cos(\theta) = \frac{y}{3}$, hence $y = 3\cos(\theta)$. Now we differentiate, and see

$$\frac{dy}{d\theta} = -3\sin(\theta)$$

To answer the other questions, note that θ never gets larger than $\pi/2$, since at that point the ladder is lying on the ground. When $0 \leq \theta \leq \pi/2$, the smaller θ gives the smaller rate of change (in absolute value); so the top of the ladder is sinking slowly at first, then faster and faster, fastest just as it hits the ground.



Suppose a point in the plane that is r centimetres from the origin, at an angle of θ ($0 \leq \theta \leq \frac{\pi}{2}$), is rotated by $\pi/2$ radians. What is its new coordinate (x, y)? If the point rotates at a constant rate of a radians per second, when is the x coordinate changing fastest and slowest with respect to θ?



Q

ans

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$$x = r\cos\left(\theta + \frac{\pi}{2}\right) = -r\sin(\theta)$$

and

$$y = r\sin\left(\theta + \frac{\pi}{2}\right) = r\cos(\theta)$$

To find how fast x is changing with respect to θ, we take $x'(\theta) = -r\cos(\theta)$. We see that when $\theta = 0$, x changes a lot when θ changes; and when $\theta = \pi/2$, x only changes a little when θ changes.

Included Work

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